

- External Gills: marine annelids
- Internal Gills: Fish, shrimps, prawns
- Tracheal systems: Insects
- Lungs: Mammals (Human), Reptiles, Birds.
- Skin: Amphibians
- Book lungs: Spiders, Scorpions

Human Respiratory System

Gross structure and function of the human respiratory system

Human respiratory system consists of the following major parts: nostrils, nasal cavity, pharynx, larynx, a series of branching ducts starting from the trachea, two bronchi one leading to each lung and smaller bronchioles, which finally end in air sacs called alveoli. The bronchioles and the alveoli are contained within the paired, cone-shaped lungs located in the thoracic cavity. The two lungs differ slightly in shape and size. The left lung is slightly smaller than the right because the apex of the heart is slightly to the left of the median plane and it has 2 lobes while the right lung has 3 lobes. Each lung is surrounded by two membranes. The inner membrane, called the visceral pleura adheres to the outer surface of the lungs while the outer membrane called the parietal pleura adheres to the wall of the thoracic cavity. Between these two membranes there is a thin, fluid filled space.

During respiration, air enters the respiratory system through the nostrils. In the nasal cavity air is filtered by hairs and is warmed and humidified as it travels through spaces in the nasal cavity. The nasal cavity leads to the pharynx which is a common passage for both air and food. That means air passage and food passage cross each other. During swallowing of food the larynx moves upwards which allows the epiglottis to close the opening of the larynx called the glottis. This allows food to go down the esophagus to the stomach. The rest of the time the glottis is open so that air can move from the pharynx through the larynx to the trachea. In the larynx are vocal cords which are made up of largely elastic fibers. These vocal cords help produce sound when expired air rushes across the stretched or tensed vocal cords, causing them to vibrate. Both the walls of the larynx and trachea are strengthened by cartilage that help these airways to keep open. The air passes from the trachea into the two bronchi that lead into each lung. Within the lungs the air passes through smaller and smaller branches of the bronchi called bronchioles.

The epithelium found in the major branches of this respiratory tract has cilia and a thin film of mucus. The mucus helps trap the dust and other particulate contaminants in the inhaled air. Then the beating of cilia moves this mucus upwards towards the pharynx where it is swallowed into the esophagus. This process is referred to as the “mucus escalator”. It helps clean the respiratory system.

At the tips of tiniest bronchioles are large number of small air sacs clustered together. The air passes into these air sacs called alveoli where gas exchange occurs. The walls of the alveoli are made up of a single layer of flattened epithelial cells that lack cilia. This inner lining of the alveoli is coated by a thin film of fluid. The lungs contain millions of alveoli. This allows a large surface area for gas exchange. Each alveolus is also surrounded by a network of capillaries. Oxygen in the air that enters the alveoli dissolves in the moist film and rapidly diffuse across the thin epithelium into the capillaries. Meanwhile a net diffusion of carbon dioxide occurs from the capillaries into the alveoli. Since there are no cilia in the alveoli, there are white blood cells in the alveoli to engulf foreign particles. Alveoli are also coated with a surfactant that reduces the surface tension thereby preventing the collapse of alveoli due to high surface tension.

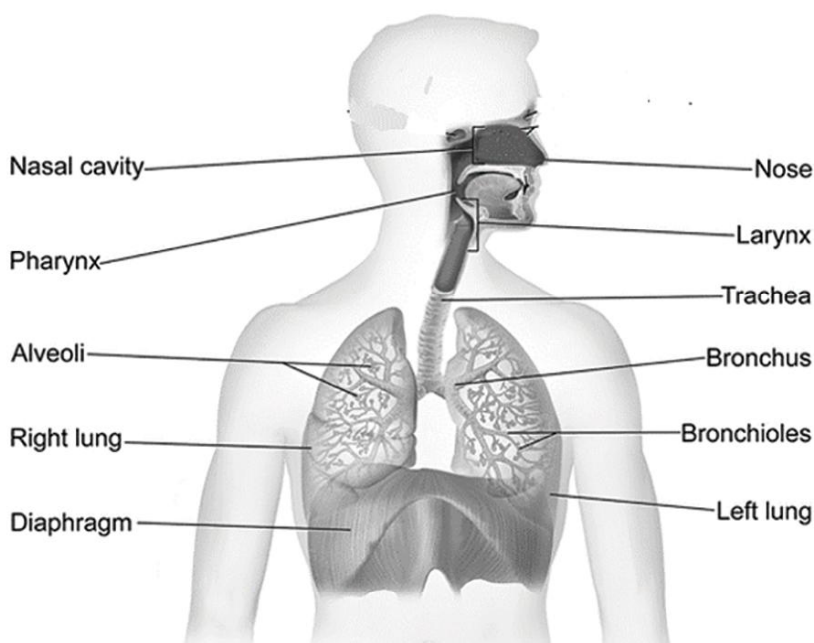


Fig 5.23: Gross structure of human respiratory system

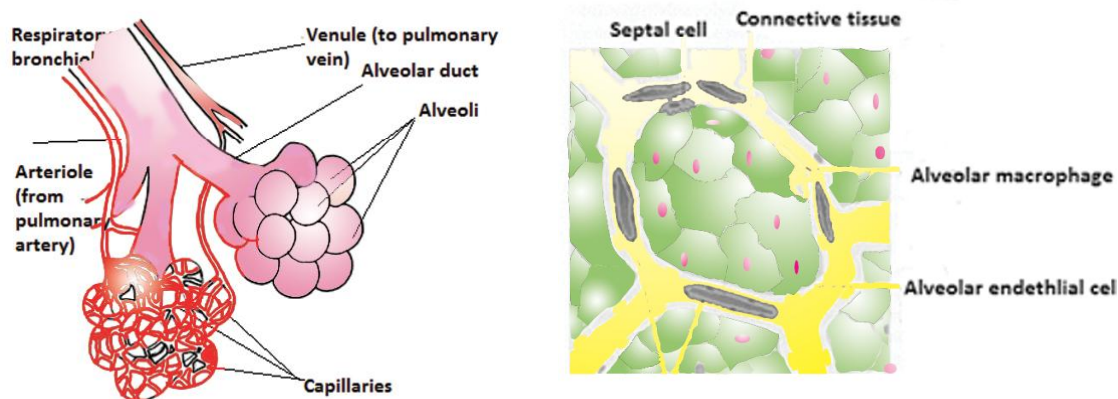


Fig 5.24: The alveoli and their capillary network

Mechanism of ventilation of the lungs

- Ventilation of the lungs is necessary to maintain high oxygen and low carbon dioxide concentrations in the alveoli or the gas exchange surface.
- Ventilation is accomplished by breathing, which is the alternating movement of air into (inhalation) and out of (exhalation) the lungs.
- Humans employ what is referred to as negative pressure breathing where air is pulled rather than pushed into lungs.
- Inhalation is an active process. Contraction of rib muscles or intercostal muscles and the diaphragm which is a sheet of skeletal muscle that forms the bottom of the thoracic cavity leads to the expansion of the thoracic cavity.
- The visceral and parietal pleurae surrounding the lung stick together due to the surface tension of the fluid between these two membranes. This allows the two membranes to slide smoothly past each other. Hence, as the volume of the thoracic cavity increases, the lung volume increases as well.
- As a result, the pressure within the lungs decrease relative to the outside air.
- This creates a pressure gradient between the atmosphere and the lungs.
- Thus, air flows from a high pressure gradient in the atmosphere to a lower pressure gradient in the lungs.
- During exhalations which is usually a passive process, the rib muscles and the diaphragm relax. This cause the volume of the thoracic cavity to reduce.
- As a result, the pressure inside the lungs increase in relation to the air outside. This pressure forces air to move out of the lungs through the respiratory tubes into the atmosphere.
- When a man is at rest contraction of rib muscles and contraction of diaphragm are enough for breathing.
- However, depending on the activity level, additional muscles may be used to aid breathing such as muscles of the neck, back and chest. These muscles further help to increase the volume of the thoracic cavity by raising the rib cage. e.g. during exercise

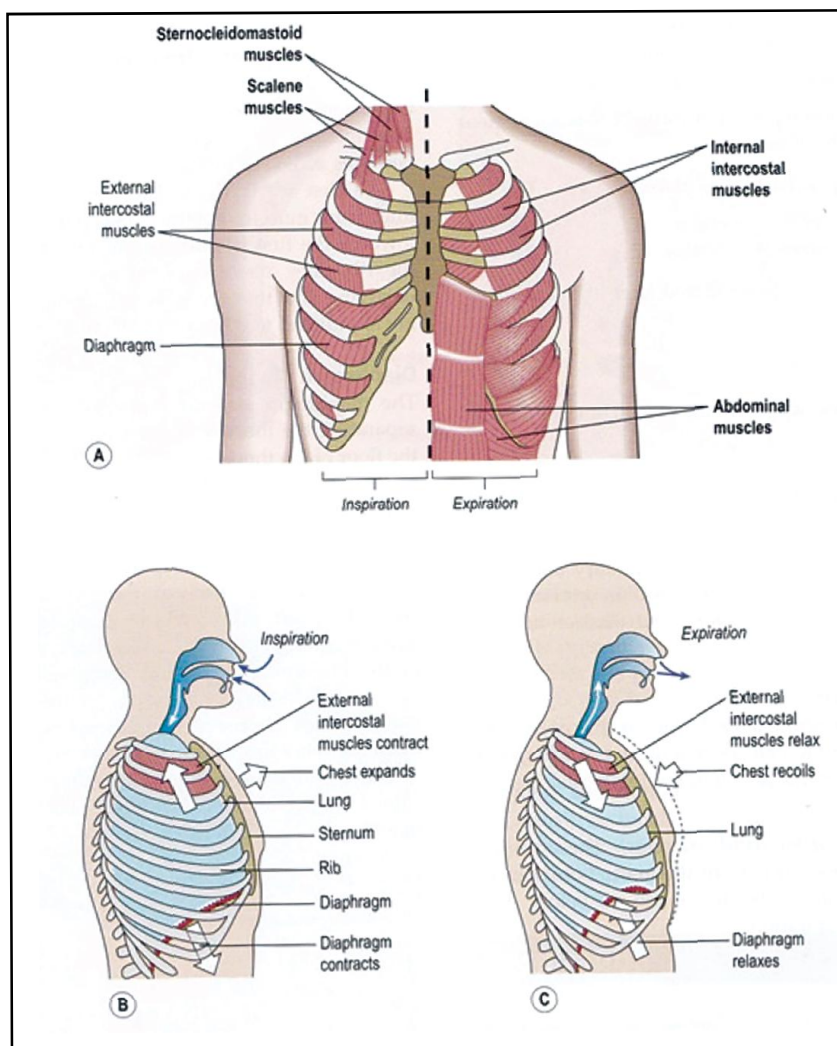


Fig 5.25: (A) Muscles involved in respiration; (B) and (C) Changes in chest volume during inspiration and expiration

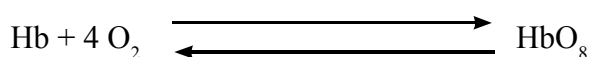
- The lungs serve as an efficient respiratory surface because:
- Alveoli create a large surface area for gas exchange.
- Alveoli and capillary walls are both lined by simple squamous epithelia which reduce the distance the gasses need to travel by diffusion.
- The alveoli surface is moist to dissolve respiratory gasses for diffusion.
- Alveoli are highly vascularized which enables the maintenance of a steep diffusion gradient of respiratory gasses

Gas exchange at the alveoli and in the tissues is a continuous process. It requires transport of O_2 from the lungs to the blood and movement of CO_2 from the blood (referred to as external respiration) and movement of O_2 from blood to the tissues and CO_2 from tissues to the blood (referred to as internal respiration).

Diffusion of O_2 and CO_2 requires partial pressure gradients between the alveolar air in the lungs and blood (during external respiration) and blood and tissues (during internal respiration).

During inhalation, fresh air mixes with the stale air in the lungs. This mixture in the lungs has a higher partial pressure of oxygen (PO_2) and a lower partial pressure of carbon dioxide (PCO_2) than the blood in the alveolar capillaries. There is thus a concentration gradient favoring the diffusion of these two gases in opposite directions. Net diffusion of O_2 takes place from the air in the alveoli to the blood and net diffusion of CO_2 takes place from the blood into the alveoli.

When O_2 molecules diffuse into blood capillaries they bind to haemoglobin in the red blood cells. Four molecules of O_2 bind reversibly with one molecule of haemoglobin and form oxyhaemoglobin.



When blood leaves the alveolar capillaries the oxygen and carbon dioxide partial pressures are in equilibrium with those of alveoli air. Once this blood returns to the heart, it is pumped through the systemic circuit.

Blood reaching the tissues in the systemic capillaries have a higher PO_2 and a lower PCO_2 than in the tissues. These partial pressure gradients result in the net diffusion of O_2 from the blood stream into the tissue and CO_2 diffusion from the cells into the blood stream across the extracellular fluid/interstitial fluid. This is called unloading of O_2 and loading of CO_2 . Then the blood returns to heart and pumped to lungs again.

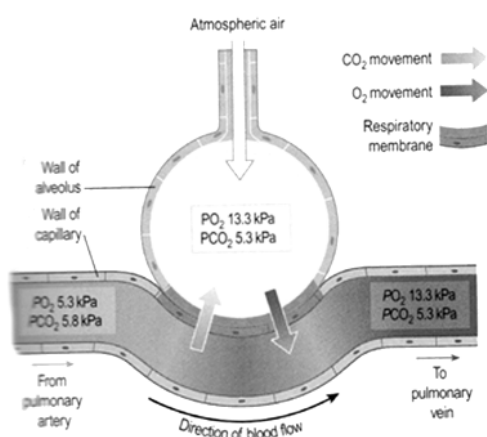


Fig 5.26: External respiration

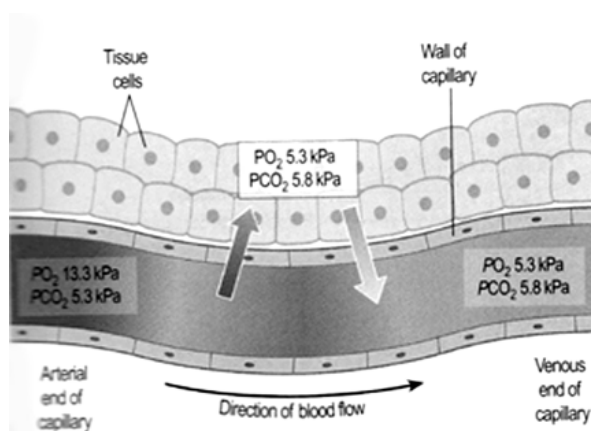


Fig 5.27: Internal respiration

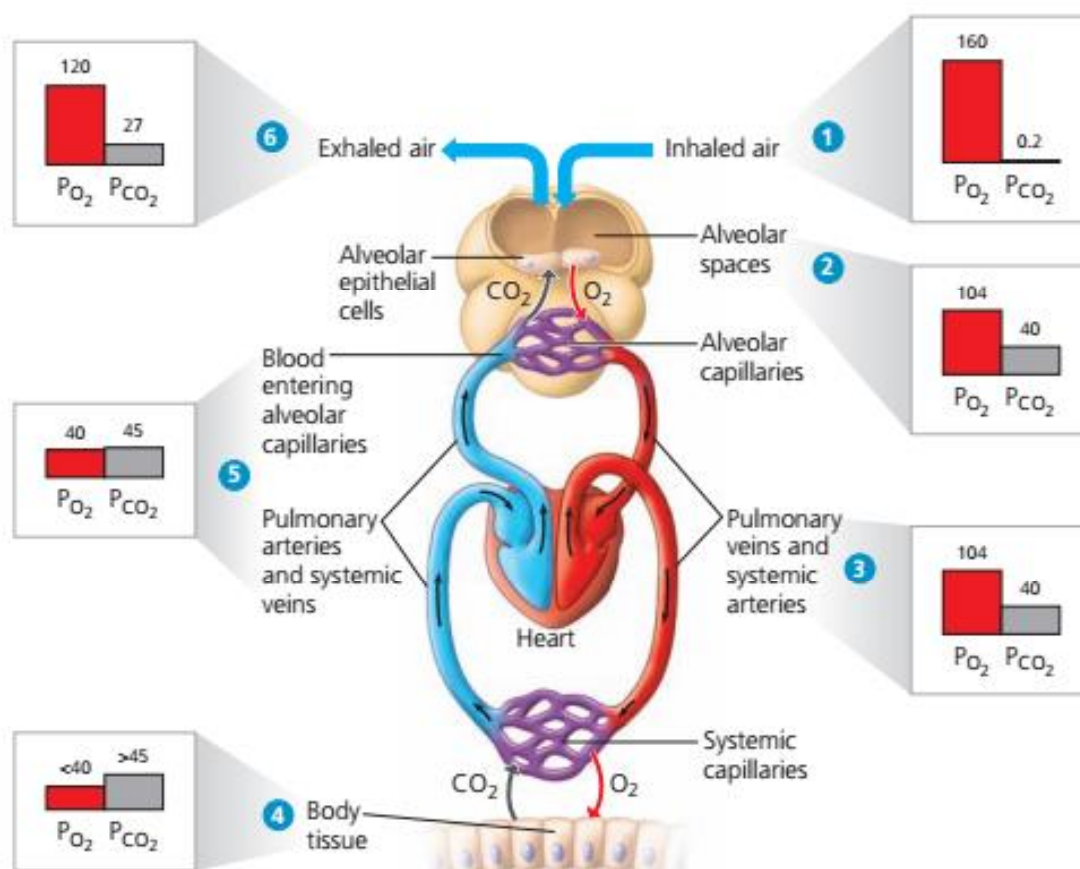


Fig 5.28: Loading and unloading of respiratory gases

Homeostatic control of Breathing

- Typically breathing is regulated by involuntary mechanisms. These involuntary mechanisms help coordinate gas exchange with blood circulation and metabolic demands
- Medulla oblongata is the main breathing regulating center found at the base brain. There are a pair of breathing control centers found in medulla and they are responsible for regulating the breathing rhythm.
- A negative-feedback mechanism is involved in regulating this process. Sensors which detect stretching of the lung tissues are found in the lungs. During inhalation, these sensors send nerve impulses to the neurons that act as control circuits in the medulla and further inhalation is inhibited and this prevents the lungs from over expanding.
- To regulate breathing, the medulla depends on pH changes in tissue fluids. The pH of tissue fluid is an indicator of blood carbon dioxide concentration. For

example, when metabolic activities increase, the concentration of CO_2 in the blood increases. Because CO_2 diffuses into the cerebrospinal fluid, this results in an increase of CO_2 concentration in the cerebrospinal fluid as well. There CO_2 reacts with water and forms carbonic acid (H_2CO_3). H_2CO_3 dissociates into HCO_3^- and H^+



- Hence, a high CO_2 concentration results in an increase in H^+ concentration, thereby a lower pH.
- This pH change is detected by the sensors in the medulla and in major blood vessels called arteries and aorta.
- Sensors in the medulla and major blood vessels detect this decrease in pH. In response, the control circuits in medulla increase the depth and rate of breathing until the excess CO_2 is removed in exhaled air and the pH of blood comes to its normal value which is 7.4
- The O_2 level has little influence on the breathing control centers. But, when O_2 concentration becomes very low, O_2 sensors found in the aorta and the carotid arteries send impulses to the medulla to increase the breathing rate.
- The regulation of breathing is also modulated by additional neural circuits in the pons, a part of the brain stem found above the medulla.

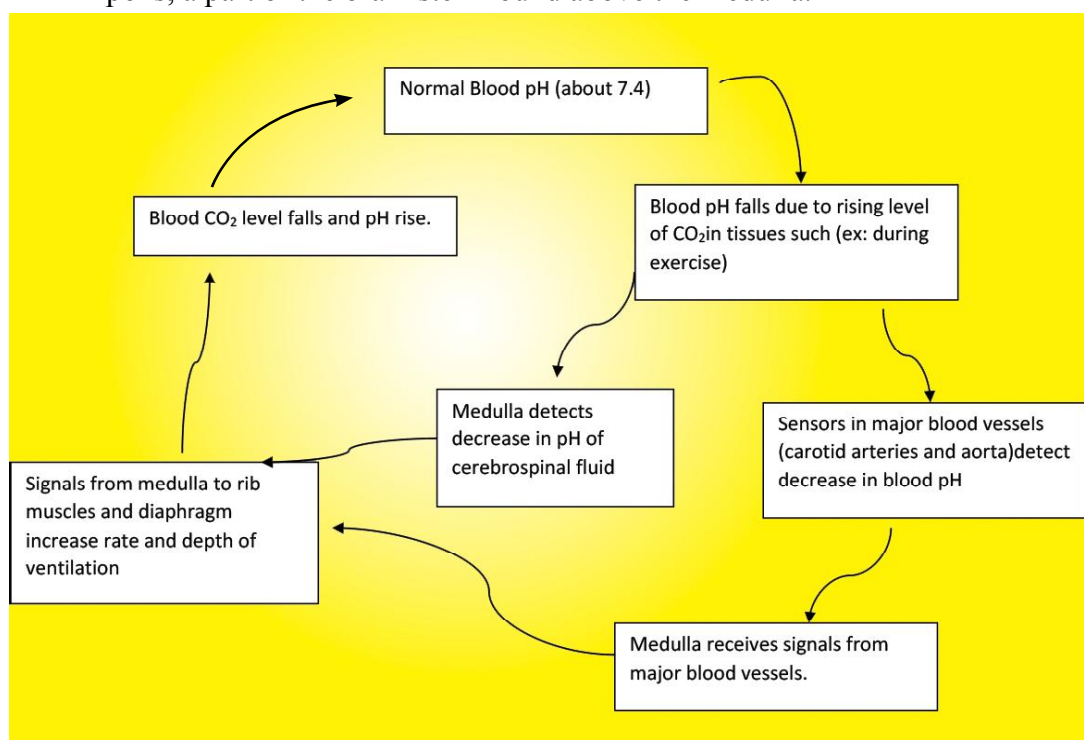


Fig5.29: Homeostatic control of breathing

Disorders of the respiratory system**The effect of smoking on the smooth functioning of the respiratory system**

- Cigarette smoke harms nearly every organ in the body including the organs of the respiratory system and increase the risk of illness, disability and death. Smokers inhale large number of chemicals which mainly come from burning tobacco. Some of these inhaled compounds are chemically active and can trigger damaging changes in the body.
- Nicotine is among the compounds inhaled in tobacco smoke which is the addictive drug in the smoke. It temporarily increases the rate of heart beat and constriction of peripheral blood vessels causing a temporary increase in blood pressure.
- Cigarette smoke stimulates the secretion of mucus by the goblet cells and inhibits the action of cilia in the respiratory tract causing accumulation of mucus in bronchioles and blocking them, leading to bronchial inflammation or bronchitis. As a result, breathing may become difficult.
- Some chemicals such as hydrogen cyanide in cigarette smoke stops the cilia from working properly. Due to loss of action of cilia, dust and other particulate matter get collected in the lung, resulting in an increase in phagocytic cells in the lung tissue. Due to release of large amounts of lytic enzymes by these cells, the alveolar tissue is destroyed thus reducing the effective area available for gas exchange.
- Carbon monoxide (CO) present in tobacco smoke is absorbed into the blood and is able to bind to hemoglobin better than oxygen and combines irreversibly with hemoglobin. Thus it decreases the amount of oxyhaemoglobin produced. Therefore, oxygen transport through blood is decreased.
- Tobacco smoke also contains a large number of cancer-causing substances (carcinogens). Nearly 90% of lung cancers are due to smoking. Long term exposure to such chemicals in cigarette smoke results in the proliferation of cells in the bronchial epithelium, forming a mass of abnormal cells. A cancer may develop among these cells. If these cells break free, the cancer may spread to other parts of the lungs and or to other organs.
- Passive or second hand smoking will also result in the above mentioned ill effects.

Silicosis

This may be caused by long-term exposure to dust containing silica compounds. High risk industries are,

- Quarrying granite, slate, sandstone

- Mining hard coal, gold, tin, copper
- Stone masonry and sand blasting
- Glass and pottery work

When silica particles are inhaled they accumulate in the alveoli. These particles are ingested by macrophages, some of which remain in the alveoli and come out in to the connective tissue around bronchioles and blood vessels close to the pleura. Progressive fibrosis is stimulated which eventually obliterates the blood vessels and respiratory bronchioles. Gradual destruction of lung tissue eventually leads to pulmonary hypertension and heart failure.

Asbestos related diseases - Asbestosis

Those who are involved in making or using products containing asbestos are at risk. This occurs when asbestos fibers are inhaled with dust. In spite of their large size the particles penetrate the level of respiratory bronchioles and alveoli. Macrophages accumulate in the alveoli and the shorter asbestos fibers are ingested. The larger fibers are surrounded by macrophages, protein materials and iron deposits. The macrophages that have engulfed fibers move out of the alveoli and accumulate around respiratory bronchioles and blood vessels, stimulating the formation of fibrous tissue. These cause progressive destruction of lung tissue and pulmonary hypertension.

Lung cancer

Nearly 90% of lung cancer is due to smoking. When one smokes, the nasal hairs, mucus and cilia in the respiratory tract that otherwise is sufficient to protect the lung from chemical and biological irritants, are overwhelmed and eventually stop functioning. As a result, irritants, free radicals, carcinogens and pathogens accumulate in the lungs. Eventually these cause lung cancer.

Tuberculosis (TB)

Tuberculosis is an infectious disease caused by the bacterium *Mycobacterium tuberculosis*. The bacterium spreads when an infected person coughs and the bacterium enters the body of an uninfected person through inhaled air. The most common form is Pulmonary TB which affects the lungs. Other organs may also be affected. Transmission of pulmonary TB is by inhaling the bacterium into the lungs. This bacterium can survive in the air and in the house dust for long periods. Malnutrition and other infectious can reduce resistance to the disease.

Symptoms

Loss of appetite, loss of weight, excessive sweating, fever, a racking cough and spitting up blood.

Asthma

Asthma is characterized by wheezing and chest tightness causing breathing difficulty. This is caused by the sudden contractions of smooth muscles in the walls of the bronchioles which causes the bronchioles to narrow or even close. During this time breathing causes whistling or wheezing sound. The cause of asthma is an over reaction of an immune response to stimuli like pollen, dust, mites, spores, particular food, cold air, exercise, smoking gases. Anti-inflammatory drugs help control.

Respiratory cycle and lung volumes and capacities

Inhalation and exhalation during a single breath is referred to as a respiratory cycle. The amount of air that flows in and out of the lungs depends on the conditions of inspiration and expiration. Thus, four respiratory volumes are described.

- **Tidal volume (TV):** This is the volume of air inhaled and exhaled with each breath during normal breathing. On average it is about 500 ml in a resting adult human.
- **Inspiratory reserve volume (IRV):** This is the extra volume of air that can be forcibly inhaled beyond the tidal volume.
- **Expiratory reserve volume (ERV):** The extra volume of air which can be expelled from the lungs after a tidal expiration.
- **Residual volume (RV):** The volume of air that remains in the lungs even after forceful expiration. This is on average is about 1,200 ml.

Specific combinations of respiratory volumes are called respiratory capacities. Thus, respiratory capacities always consist of two or more lung volumes. The respiratory capacities are important to determine the respiratory status of a person.

- **Inspiratory capacity (IC):** The total volume of air that can be inspired after a tidal expiration.

Thus, $IC = TV + IRV$

- **Functional residual capacity (FRC):** The volume of air remaining in the lungs at the end of a tidal expiration.

Thus, $FRC = RV + ERV$

- The functional residual capacity is important for continuous exchange of gas in the alveoli and to prevent the collapse of the alveoli during expiration.
- **Vital capacity (VC):** The maximum volume of air which can be inhaled and exhaled. It is normally around 3100 mL in women and 4800 mL in men.

$VC = TV + IRV + ERV$

- **Total lung capacity (TLC):** The maximum volume of air the lungs can hold or the sum of all lung volumes. This is normally around 6000 mL.

In addition, some of the inspired air fills the system of branching conducting tubes (trachea, bronchi and bronchioles) and never contributes to the gas exchange in the alveoli. This volume is referred to as the anatomical dead space and it is typically about 150 mL.

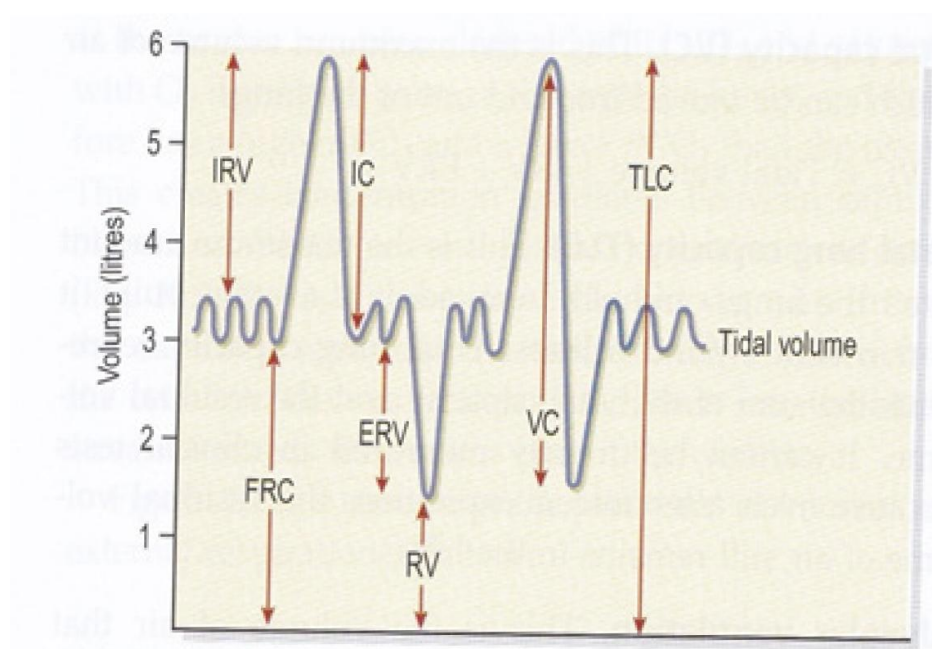


Fig 5.30: Lung volumes and capacities

- IRV: Inspiratory reserve volume
IC: Inspiratory capacity
FRC: Functional residual capacity
ERV: Expiratory reserve volume
RV: residual volume
VC: Vital capacity
TLC: Total lung capacity

Immunity

The state of being resistance to injury, invading pathogens and foreign substances through defensive mechanisms in the body is known as immunity. Pathogens which are the agents that can cause infectious diseases include some bacteria, virus, and fungi. The substances that can be recognized as foreign to the body include chemical components of pollen grains, incompatible blood cells and transplanted tissues. Defenses in the animal body against these foreign invasions make up the immune system. Immune system of animals should be able to distinguish own body cells (self) from foreign cells/particles within the body (non-self) for initiating defensive immune responses for destruction of the foreign agents such as pathogens. Special cell types in the body fluid and tissues of most animals can interact with these foreign invasions and destroy them. Immune cells produce receptor molecules that bind specifically to molecules from foreign cells and agents and activate defense responses.

Immune responses in animals can be divided into two types:

- Innate immunity
- Acquired immunity (Adaptive immunity)

Innate immunity

Innate immunity is the ability to resist damage or diseases in the body through inherent body defenses which offer rapid responses against a broad range of pathogens and foreign substances. In innate immunity, recognition and defense responses will depend on characters common to groups of pathogens. Innate immunity lacks specific responses to specific invaders and its protective mechanisms functions the same way regardless of the type of the invader. Hence innate immunity is also known as non-specific defense. Innate immune responses include defense mechanisms that provide immediate but general protection against foreign invasion. Innate immunity is found in both invertebrates and vertebrates. Innate defense mechanisms can be broadly divided into two types: External barriers (External defenses/ barrier defense) and Internal nonspecific defenses (Internal defenses).

External defenses/ barrier defense in innate immunity

External barriers discourage pathogens and foreign substances from penetrating the body. So they are considered as the first line of defense. For innate immunity in the human body, external defenses/barriers are found in the skin, mucus membranes and secretions of various organs. They act as physical and chemical barriers.

- Human **skin** with its many layers of closely packed, keratinized cell layers in the epidermis provides a significant physical barrier to entrance of microbes. In addition periodic shedding of epidermal cells helps remove microbes from the skin surface.
- The **mucous membranes** which line the body cavities provide a physical barrier to entrance of many microbes (e.g. the linings of the respiratory tract, digestive tract, urinary tract and reproductive tract). The mucous membranes produce mucus which traps microbes and other particles. In the respiratory tract, ciliated epithelial cells sweep mucus and any entrapped material upward. Coughing and sneezing speed up the mucus movement and its entrapped pathogens out of the body preventing their entry to the lungs.
- **Secretions** by various organs (e.g. tears, saliva, mucus) help as physical and chemical barriers to protect epithelial surface of the skin and mucous membranes. Tears in the eyes provide protection against irritants and microbes. Tears in the eyes provide continuous washing action that helps to dilute microbes and prevent settling on the surface of eyes. Saliva washes microbes from the mouth surface and the flow of saliva reduces the colonization of microbes in the mouth. Mucus secretions which bathe various exposed epithelia provide a continual washing action to dilute and inhibit colonization microbes such as bacteria and fungi. Lysozyme (an enzyme) present in tears, saliva, perspiration and mucous secretions can destroy cell walls of some bacteria. Gastric juice which provides an acidic environment in the stomach can destroy many bacteria and bacterial toxins ingested with food. Secretions of the sweat and sebaceous glands of the skin give acidity of the skin which helps to prevent growth of bacteria.

Internal defenses in Innate immunity

When the pathogens penetrate the external defensive barriers in the skin and mucus membranes in the human body, they encounter a second line of innate immunity responses called internal defenses. Within the body, detection of non-self is accomplished by molecular recognition in which receptor molecules on specific cells in the immune system will bind specifically to molecules of foreign agents such as pathogens.

In innate immunity, internal defenses consist of Phagocytic cells, Natural killer cells, Antimicrobial proteins and Inflammatory responses.

- **Phagocytic cells:** These are specialized cells that can ingest microbes, foreign particles and cell debris for intra cellular digestion and destruction. Phagocytes use the receptor molecules to detect components of foreign agents and particles. Neutrophils and Macrophages are the two main types of phagocytic cells in man. While circulating in the blood, neutrophils are attracted first to the infected site

by signals from affected tissues. Then neutrophils can ingest and destroy infected pathogens. Macrophages are larger and more potent phagocytic cells.

- **Natural killer cells:** These are a type of lymphocytes present in the blood and some tissue/organs such as spleen and lymph nodes which function in nonspecific defense. They can detect the cells with abnormal surface molecules (e.g. virus-infected body cells and some cancerous cells) and kill them. Natural killer cells do not engulf these abnormal cells but upon binding they can release chemicals to kill the virus-infected cells and cancerous cells which could inhibit further spread of the virus or cancer.
- **Antimicrobial proteins:** They are proteins present in the blood and interstitial fluids which function in innate defense by attacking microbes directly or impeding their reproduction. Interferons and Complement proteins are two such antimicrobial proteins that discourage microbial growth. **Interferons** are proteins secreted by virus-infected body cells that protect uninfected host cells from viral infections by interfering with the viral replication. Once released by virus-infected cells, interferons diffuse to uninfected neighboring cells where they are stimulated to produce “anti-viral proteins” which inhibit viral replication. Some interferons activate macrophages which enhance the phagocytic activity. Complement proteins are a group of normally inactive proteins in the blood plasma and plasma membranes. When they are activated by different substances present on the surfaces of microbes, a cascade of biochemical reaction occurs which lead to lysis of invaded cells. They also promote phagocytosis and inflammatory response.
- **Inflammatory response:** This is an innate immune defense response in the body to tissue damage triggered by microbial infections or injury of the tissues. This involves the release of substances that promote increased permeability and dilation of blood vessels, enhance migration of phagocytes, destruction of invading pathogens and aid in tissue repair (Figure 5.31). Inflammation attempts to destroy the microbes at the site of the injury preventing the spread to other tissues and promote tissue repair.

Inflammatory response is brought about by various signaling molecules upon infection or injury. Histamine is one of the important inflammatory signaling molecules released mainly by mast cells in the connective tissues at the site of damage. Histamine causes increased permeability and dilation of nearby blood vessels (blood capillaries). Increased permeability of the blood vessels enhance the infiltration of white blood cells, antimicrobial proteins and clotting elements to enter the injured area from the blood that aid in destruction of invading pathogens and tissue repair. Dilation of blood vessels allows more blood to flow through the damage area which helps to remove dead cells. Activated phagocytes (macrophages and neutrophils) moved from the

blood to the damaged tissue area can also discharge signaling molecules (cytokines) which also promote blood flow to the injured or infected site. During inflammation, activated complement proteins can cause further histamine release which attracts more phagocyte cells to enter injured tissue and carry out additional phagocytosis. This process can digest the microbes and cell debris at the site of injury.

Signs and symptoms of inflammation are redness, heat, swelling and pain. Dilation of blood vessels causes redness and heat production due to high metabolism in the area. Increased permeability of blood vessels triggers localized swelling due to leaking of tissue fluid into neighboring tissues. Pain results from injury to neurons and microbial toxins. As a result of most inflammatory responses, pus may be accumulated. It is a fluid rich in dead phagocytes, dead pathogens and cell debris from the damaged tissue. Minor injury or infection causes a localized inflammatory response. If the injury or infection is severe it may lead to a systemic response (throughout the body) leading to fever. Elevated body temperature within limits may enhance the phagocytosis and accelerate tissue repair by speeding up the chemical reactions.

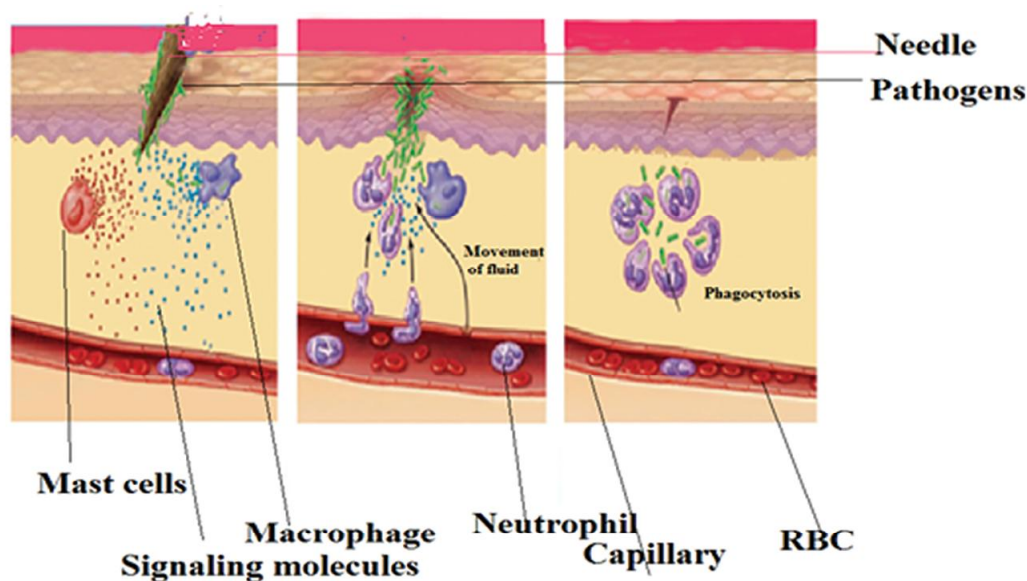


Fig 5.31: Major events in inflammatory response

Acquired Immunity (Adaptive Immunity)

Acquired immunity is the ability of the body to defend itself against invading foreign agents (e.g. pathogens) through specific defense responses mediated by diverse **T lymphocytes and B lymphocytes**. Acquired immunity shows (i) specificity for particular foreign molecules (ii) recognition of animals own molecules (self-molecules) from non-self-molecules and (iii) memory for most previously encountered pathogens such that the subsequent encounter causes a stronger and more rapid response (immunological memory). In the animal kingdom, acquired immunity is found only in the vertebrates.

The cells that have developed the ability to carry out acquired immune responses if they are activated against foreign agents are called **T lymphocytes** and **B lymphocytes**. In man, both types of lymphocytes are originated from stem cells in the bone marrow. Some of the lymphocytes that migrate to thymus for maturation are called T lymphocytes (T cells). The lymphocytes that remain in the bone marrow for completion of development are called **B lymphocytes** (B cells). Before leaving these lymphocytes to the secondary lymphatic tissues, their plasma membranes acquire diverse specific protein receptors (antigen receptors) which have the ability to recognize specific foreign invasions (There can be over 100,000 antigen receptors on the surface of a single B lymphocyte or T lymphocyte)

An **antigen** is a substance that has the ability to stimulate an immune response through T lymphocytes and B lymphocytes and to react with the specific cells or antibodies that resulted from the stimulated immune response. Viral proteins, bacterial toxins and chemical components of bacterial structures such as flagella and cell walls can be antigenic. Structural components of incompatible blood cells, transplanted tissues can also be antigenic. Antigens are usually large foreign molecules such as proteins and polysaccharides. In general not the entire antigen, but certain parts of a large antigen molecule act as the triggers for the acquired immune responses. The small accessible portion of the antigen that binds to a specific antigen receptor of a T lymphocyte or B lymphocyte is called an epitope (for exp a group of amino acids in a large protein). Can serve as an epitope. Usually a single antigen has several epitopes (Figure 5.32) each can bind with a specific antigenic receptor of the single T or B lymphocyte.

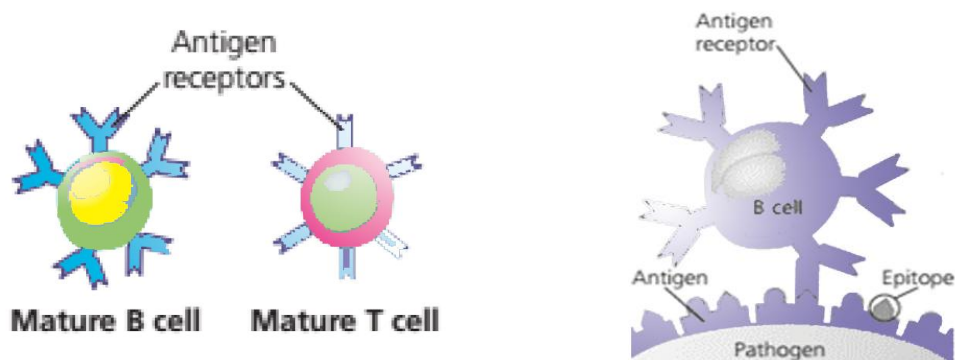


Figure 5.32: (a). Mature T lymphocyte and mature B lymphocyte with antigen receptors on the plasma membrane, (b). B cell receptor binds with an Epitope on antigen

In acquired immunity, two types of immune responses are mediated by T lymphocytes and B lymphocytes. They are Cell mediated immune responses and Humoral immune responses. Humoral immune response is also called as Antibody mediated immune response. Both immune responses are triggered by antigens. A given pathogen may provoke both types of immune responses.

Cell mediated immune response

Cell mediated immune response is a type of acquired immunity in which specifically sensitized T lymphocytes attach to the antigen undergo proliferation and eventually differentiate into “Cytotoxic T cells” that can directly kill the cells with the invading antigen. In addition “Memory T cells” are formed that can cause stronger and more rapid response at the subsequent encounter of the same antigen to the body. This is particularly effective against infected cells (fungi, parasites and virus that are present within host cells), some cancer cells and foreign transplanted cells. Cell mediated immunity always involves cells attacking cells.

Humoral immune response

Humoral immune response is a type of acquired immunity in which specifically sensitized B lymphocytes attach to a particular antigen undergo proliferation and eventually differentiate into “Plasma cells” that secrete circulating antibodies that can neutralize and inactivate the specific toxins and pathogens in the blood and lymph. In addition “Memory B cells” are formed that can cause stronger and more rapid response at subsequent encounter of the same antigen. Humoral immune response works mainly against antigens present in body fluids and extracellular pathogens (mainly bacteria) that multiply in the body fluids.

Antibodies

Antibodies are proteins secreted by plasma cells (differentiated B lymphocytes) in response to specific antigens; the antibody binds with that antigen to neutralize, inhibit or destroy it. Antibodies can neutralize and inactivate the specific toxins and pathogens in the body fluids. The antibodies do not directly kill the pathogens but can interfere with activity of the pathogen or mark the pathogen for inactivation and destruction. Antibody-antigen complexes can activate complement system and phagocytosis to destroy the pathogen. Antibodies are also called as immunoglobulins. Immunoglobulin has the same Y shaped structure as B lymphocyte antigen receptors but are secreted than membrane bound.

Role of T lymphocytes and B lymphocytes in acquired immunity**Recognition of the antigen, binding to the antigen and sensitization:**

For an acquired immune response to occur, some T lymphocytes or B lymphocytes must first recognize that a foreign antigen is present in the body. Even though there are vast variety of antigen receptors present on different B lymphocytes and T lymphocytes, only a very small fraction are specific for a particular epitope. Hence, antigen should be presented to the B lymphocytes and T lymphocytes until a match is made. Recognition of the antigen occurs through successful match between an epitope of the antigen and an antigen receptor on small number of B lymphocytes or T lymphocytes. As specific antigen receptors produced by a single T cell or B cell can be identical they can bind to the same epitope. Hence, both T and B cells can respond to any pathogen that produces molecules containing that same epitope. But B cells and T cells encounter antigens in different ways. T lymphocytes only recognize the fragments of antigenic proteins that are presented to the cells by a special cells called “antigen presenting cells” (macrophages, dendritic cells and B cells). However, B lymphocytes can recognize and bind to the antigens present in blood plasma, lymph and interstitial fluid. The binding of an antigen to the specific antigen receptor results in sensitization (activation) of a specific T lymphocyte or B lymphocyte which initiates cell mediated and antibody mediated immune response as described below.

Proliferation and differentiation into Effector cells: Once activated the T lymphocyte or B lymphocyte undergoes multiple cell divisions (proliferations) resulting a clone, a population of cells that are identical to the original lymphocyte. Some cells of clones become Effectors cells which are short lived cells that take effect immediately against antigen to provide primary immune responses.

Elimination of invaders: The effector forms of T lymphocyte are “Cytotoxic T cells” and “Helper T cells”. Cytotoxic T cells use toxic proteins to kill the cells infected with the pathogen. Signals from Helper T cells activate cytotoxic T cells to kill the infected cells. Signals from Helper T cells can also activate B lymphocytes to initiate antibody production. Effector forms of B lymphocytes are “Plasma cells”. A single activated B lymphocyte can form thousands of identical Plasma cells. The plasma cells begin producing and secreting a soluble form of the B lymphocyte antigen receptor (antibodies) in large quantities which are released to the blood and lymph. Hence, circulating antibodies can neutralize and inactivate the specific toxins and pathogens in the body fluids.

Provide immunological memory: Following differentiation into Effector T cells (Cytotoxic T cells and Helper T cells), other T lymphocytes in the clones remain as “Memory T cells” which are long lived that can give rise to Effector T cells if the same

antigen is encountered later in the life. Similarly the remaining B lymphocytes in the clones are “Memory B cells” which are long lived that can give rise to Plasma cells if the same antigen is encountered later in the life. These Memory T cells and Memory B cells can cause stronger and more rapid response at subsequent encounter of the same antigen to the body. This immunological memory is called secondary immune responses.

Active immunity

Active immunity is a long lasting immunity mediated by the action of B lymphocytes and T lymphocytes in the body and the resulting B and T memory cells specific for a pathogen. Active immunity can be developed as a result of natural infection of a pathogen or artificial immunization.

Naturally acquired active immunity

Long lasting immunity developed in the body against various infectious diseases in response to natural infections of pathogens is called naturally acquired active immunity. In response to a disease causing agent entering the body naturally for the first time (e.g. Virus of Chickenpox), some T lymphocytes and B lymphocytes in the body become activated and eventually produce specific cytotoxic T cells and antibodies to destroy the pathogen. Memory B cells and T cells produced in this process are long lived that will provide a stronger and rapid immune responses to destroy the particular antigen if the same antigen (e.g. Virus of Chickenpox) is encountered later in the life. In this way the body can resist to subsequent infections of the same antigen.

Artificially acquired active immunity

Long lasting immunity induced artificially in the body against various infectious diseases through vaccination (immunization) of attenuated (virulence-reduced) pathogens is called artificially acquired active immunity. Immunization can be carried out with preparations of antigens (vaccines) from many sources such as killed or weakened pathogens, inactivated bacterial cells or genes encoding microbial proteins. These vaccines act as the antigens and stimulate cell mediated and antibody mediated immune responses leading to production of long lived memory B and T cells to destroy the antigen. If the pathogen from which the antigen was derived, is encountered naturally later in the life, long lived memory cells can provide a stronger and rapid immune responses to destroy the particular pathogen. In general, the antigens used in the vaccines are pretreated to be immunogenic but not pathogenic. For example, BCG vaccine which is used against tuberculosis disease in man, has been prepared from a strain of the attenuated live tuberculosis bacteria. Polio vaccine consists of live attenuated poliovirus strains. Polio vaccine produces antibodies in the blood against polio virus, and in the event of infection, this protects the individual by preventing the spread of poliovirus to the nervous system.

Passive Immunity

Passive immunity is the short term immunity developed within the body due to the transfer of antibodies produced by another individual. Passive immunity provides immediate protection, but the body does not develop memory as passive immunity does not involve recipients' T cells and B cells. Passive immunity persists only until the transferred antibodies last (few weeks to few months). Therefore the recipient is at risk of being infected by the same pathogen later unless they acquire active immunity or vaccination. Passive immunity can be developed as a result of transferring antibodies to the recipient naturally or artificially.

Naturally acquired passive immunity

Short term antibody mediated immunity for some infectious diseases can be developed within the body of the fetus or nursing infant due to the natural transfer of antibodies produced by the mother. The immunity occurs due to the transfer of antibodies to the fetus blood from mother's blood across the placenta. Antibodies also can pass from mother to the nursing infant through the colostrum and the milk during breast feeding. The baby develops the resistance against some infectious diseases for a short time. In this way the infant may be protected from these diseases until its own immunity system is fully functional. This is known as naturally acquired passive immunity.

Artificially acquired passive immunity

Artificially acquired passive immunity is a temporarily induced defensive protection achieved by the transfer of antibodies artificially to the blood of the recipient from another source. These readymade antibodies can be administered as blood plasma or serum (human or animal), or as injections of pooled human immunoglobulin from immunized donors or as monoclonal antibodies. Passive transfer of antibodies is used to prevent some infectious diseases when infectious agents are suspected to have accidentally entered the body (e.g. readymade human serum antibodies for hepatitis A virus). It is also used in the treatment of several types of acute infections (e.g. readymade human anti-tetanus immunoglobulin for acute conditions of tetanus). Passive immunization is also used to treat poisoning from venomous snake bite (e.g. antivenin, serum prepared from horses that have been immunized against snake venom). Immunity derived from artificially acquired passive immunization lasts for few weeks to four months.

Allergies

Some persons are overly reactive to substances that are tolerated by most other people. Antigens that induce hypersensitive reactions in some persons are called allergens. Exaggerated responses of the body to certain antigens (allergens) are called allergies. Common allergens include pollens, dust, some food (e.g. shellfish), some antibiotics

(e.g. penicillin), venom from honey bees and wasps. Whenever an allergic reaction takes place the tissue injury occurs. The most allergens stimulate production of plasma cells which secrete antibodies specific for the antigen. When the same allergen enter the body later, it become attach to the antibodies specific to the allergen which induce the mast cells to release histamine and other inflammatory chemicals. Acting on a variety of cell types these signals bring about typical allergy symptoms such as sneezing, runny nose, teary eyes and smooth muscle contractions in the airways of the lungs that can result in breathing difficulties. An acute allergic conditions sometimes lead to death of the person due to breathing difficulties and low blood pressure with a few seconds of exposure to an allergen.

Autoimmune diseases

In some persons, the immune system becomes active against particular self-molecules of the body and begins to attack the person's own tissues leading to an autoimmune disease. Possible causes for autoimmune diseases may be genetic factors, gender and unknown environmental triggers. Many autoimmune diseases affect females than males. A variety of mechanisms produce different autoimmune diseases. Some involve the production of autoantibodies that can affect normal functioning of certain body molecules. Some involve activation of Cytotoxic T cells that destroy certain body cells. Examples for autoimmune diseases include Type 1 Diabetes mellitus, Multiple sclerosis and Rheumatoid arthritis. In Type 1 Diabetes mellitus, T cells attack the insulin producing pancreatic beta cells. In Multiple sclerosis, T cells attack myelin sheaths around neurons. In Rheumatoid arthritis, painful inflammations of the cartilage and bones occur as the immune system mistakenly sends antibodies to the lining of the joints, where they attack the tissue surrounding the joints.

Immunodeficiency diseases

Immunodeficiency disease is a disorder in which responses of the immune system to antigens are defective or absent. An immunodeficiency can lead to frequent and recurrent infections and increased susceptibility to certain cancers. An inborn immunodeficiency results from a genetic or developmental defects in the production of immune system cells or specific proteins such as antibodies or proteins of the complement system. Acquired immunodeficiency can be developed later in life due to the exposure to chemicals or biological agents. Drugs used to fight autoimmune diseases or prevent transplant rejections suppress the immune system leading to an immunodeficiency state. The human immunodeficiency virus (HIV), the pathogen that cause Acquired Immunodeficiency Syndrome (AIDS) escapes and attacks the immune system of man. The HIV causes progressive destruction of immune responses in the person leading to frequent infections and increased susceptibility to certain cancers which can cause death.

Osmoregulation and excretion

Osmoregulation is processes by which organisms control solute concentrations and water balance within the body. Simple unicellular organisms such as *Amoeba*, *Paramecium* etc. use contractile vacuoles for osmoregulation. But animals have developed different structures for osmoregulation. The chemical reactions that occur in organisms result in the formation of waste products, often toxic, which must be disposed in some way. The removal of the nitrogenous metabolite and other metabolic waste products from the body is called excretion. Defecation is not considered under excretion as it involves the removal of undigested food from the gut. In many animals excretory and osmoregulatory systems are linked structurally and functionally.

Importance and need of osmoregulation and excretion

For effective body functioning and survival animals have to maintain a constant internal environment specially the relative concentrations of water and solutes within favorable limits. Therefore animals need to regulate the chemical composition of body fluids by balancing uptake and loss of water and solutes. Animal cells will swell and burst if water uptake is excessive. On the other hand, animal cells will shrink and die if water loss is high. The driving force of loss of water in animals as in plants is the concentration gradient of solutes across the cell membrane. Animals have evolved different osmoregulatory strategies depending on the environment in which they live for their survival.

Animals have to get rid of toxic products produced during metabolism in order to safeguard the composition of their internal environment. Otherwise these excretory end products become toxic to the body cells. For example protein and nucleic acids are broken down within the body cell during metabolism and the amine group is converted to ammonia which is highly toxic. Ammonia also acts as a weak base. Oxidation of glucose during metabolism will release CO_2 which is a weak acid. Accumulation of such weak acids and bases will alter the acid base balance in the internal environment. Changes in acid base balance will lead to adverse effects such as denaturation of proteins. Therefore removal of excretory products from the body is essential to maintain the internal environment within favorable limits for effective body functioning and survival.

Relationship between metabolic substrates and excretory products

Metabolic substrates in the cells are carbohydrates, fats, proteins and nucleic acids. The excretory products of these substrates will vary depending on several factors such as the chemical structure and the composition, availability of enzymes, oxygen availability and the habitat in which they live.

When carbohydrates are metabolized within the body cells when oxygen is available final excretory end products are CO_2 and water. If they are subjected to anaerobic respiration in general lactic acid is produced.

When fats are subjected to aerobic metabolism final excretory products are CO_2 and water. Since proteins contain amine groups in their structure, during metabolism of excess amino acids ammonia is produced. Since nucleic acids contain nitrogenous bases ammonia is produced as an excretory product during their metabolism. Depending on the habitat and the availability of enzymes ammonia will be further converted to other nitrogenous waste products such as urea and uric acid.

Relationship between the nitrogenous excretory products and living environment

Nitrogenous excretory products of animals are ammonia, uric acid and urea. These different forms vary significantly in their toxicity and the energy costs of producing them.

Since ammonia is highly toxic, a large volume of water is needed to excrete ammonia. Therefore typically the organisms that live in water such as bony fishes, many aquatic invertebrates and aquatic amphibians specially tadpoles excrete ammonia since they have ready access to water. Energy cost for producing ammonia for excretion is comparatively low.

Terrestrial animals do not have access to sufficient water to excrete ammonia as the main excretory product. Instead most terrestrial animals such as mammals and adult amphibians mainly excrete urea as the main nitrogenous excretory product. Urea is less toxic. However animals must expend more energy to produce urea from ammonia. Some marine fishes such as sharks also excrete urea (which they use for osmoregulation) as the main nitrogenous waste.

Some terrestrial animals such as birds, many reptiles, land snails and insects excrete uric acid as the main excretory product. Uric acid is relatively non-toxic and generally insoluble in water. Therefore it is excreted as a semisolid with trace amount of water. However uric acid production from ammonia requires more energy than urea production.

The diversity of excretory structures of animals (Fine structures are not necessary)

Body Surface - The cells of some animals which are in direct contact with the environment and eliminate excretory products by diffusion. e.g. cnidarians

Flame cells- These are specialized excretory cells connected to a network of tubule which opens to the outside of the animal. E.g. flatworms.

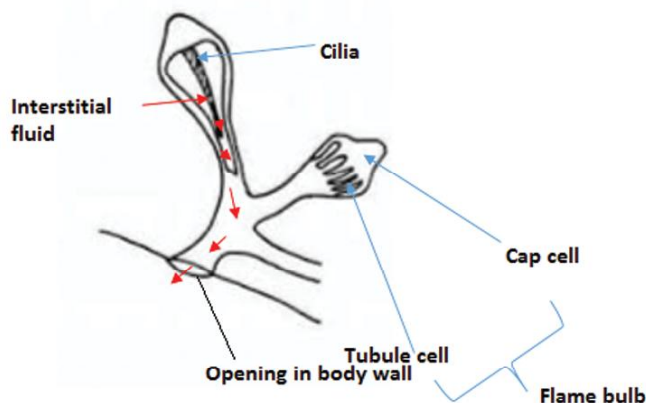


Fig 5.33: Structure of the flame cell

Nephridia – They are multi cellular, tubular structures. One end of the tubule is open to the coelom while the other end opens to the outside. e.g. Annelids

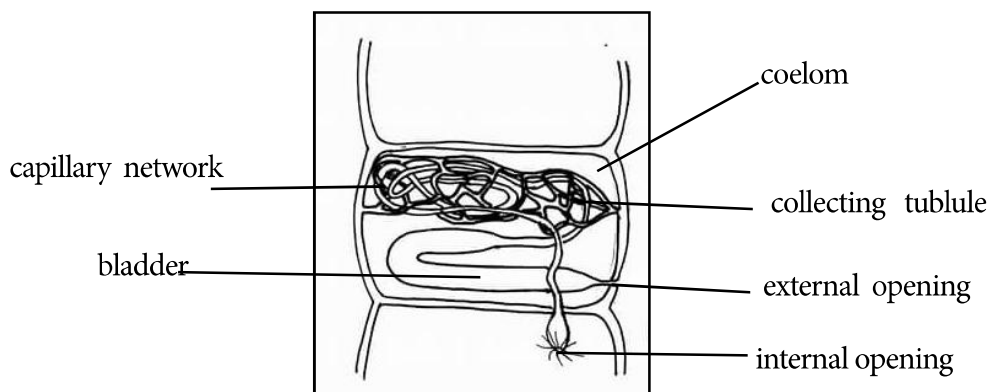


Fig 5.34: Structure of the nephridia

Malpighian tubules – These are extensive blind end tubules immersed in hemolymph and opens in to the digestive tract. e.g.: Insects and other terrestrial arthropods

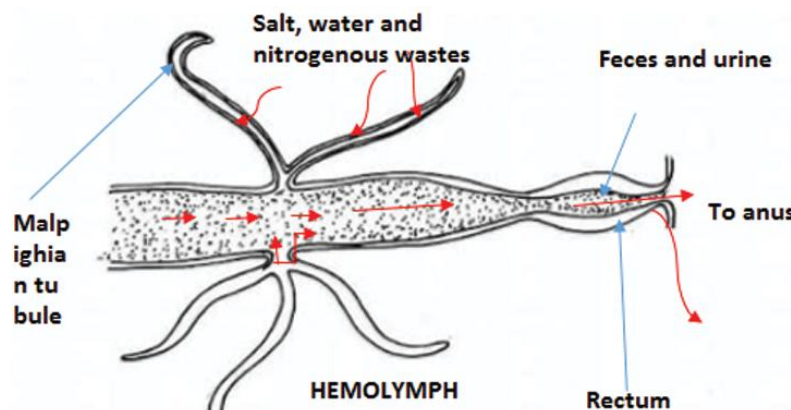


Fig 5.35: Malpighian tubules

Green glands / Antennal glands –Two large glands found ventrally in the head and anterior to the oesophagus. e.g. Crustaceans

Sweat glands –There are coiled tubular glands situated in the dermis and connected to a sweat duct which open as a pore on the surface of the skin. E.g. human skin.

Salt glands –They are paired glands found near the eyes to excrete excess salts. e.g. Marine birds and marine reptiles.

Kidney – These are the major excretory and osmoregulatory organs of all vertebrates.

Human Urinary System

Human urinary system consists of two kidneys, two ureters, urinary bladder and urethra. Their main functions are given in the following table;

Part	Main Function
Kidney	Produce urine to excrete waste products while maintaining osmotic balance and acid base balance.
Ureter	Receives urine from kidney and send it to bladder
Urinary bladder	Temporary storage of the urine
Urethra	Provide the passage through which urine stored in the bladder leaves the body

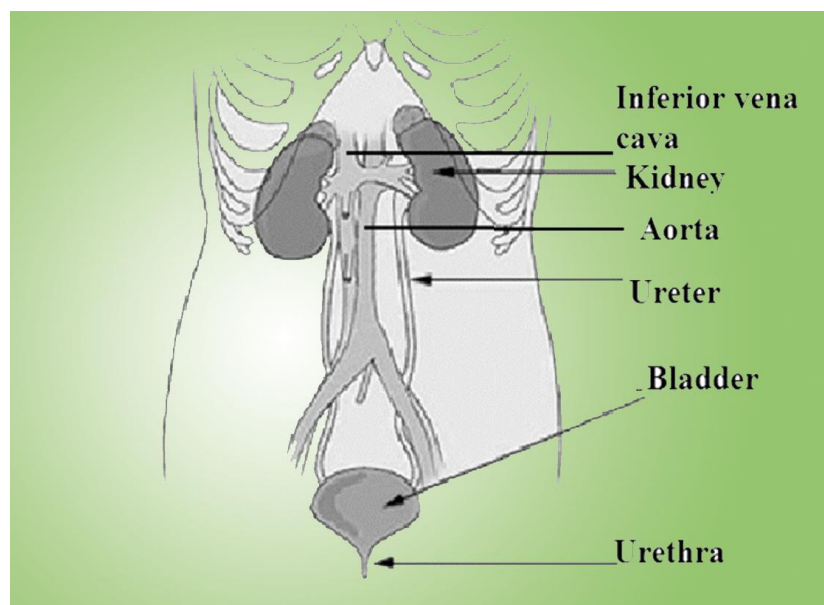


Fig 5.36: The parts of the human urinary system

Location of kidneys

Two kidneys are located on the posterior abdominal wall one on either side of the vertebral column, behind the peritoneum and below the diaphragm. Right kidney is slightly lower than the left.

Blood supply

The kidneys receive blood from aorta via the renal arteries and renal veins return blood to the inferior vena cava.

Gross structure of the kidney

Kidney is a bean shaped organ which is held in position by a mass of fat and both are surrounded by a fibrous connective tissue. In the longitudinal section of the kidney three areas of tissues can be seen to the naked eye. They are outer fibrous capsule, renal cortex and inner renal medulla. Cortex and medulla are supplied with blood vessels and tightly packed with excretory tubules. Renal Cortex is granulated due to the presence of glomeruli. Medulla is composed of renal pyramids, which have striated appearance. Apices of pyramids project in to the renal pelvis through renal papillae. Renal pelvis leads into the ureter. Renal artery and renal vein pass through the pelvis.

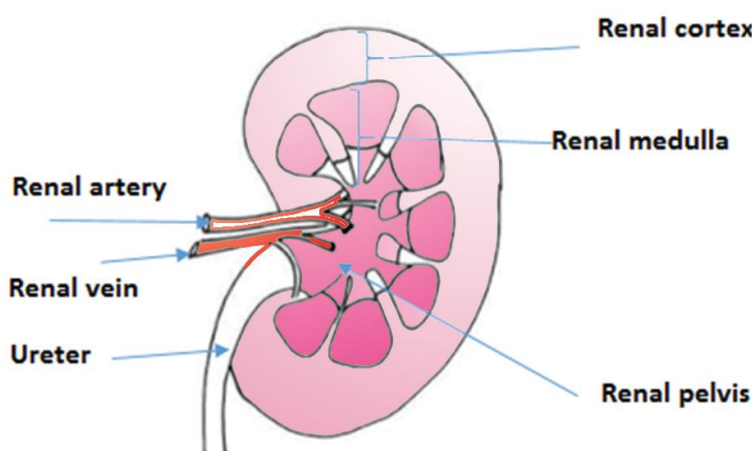


Fig 5.37: Longitudinal section of human Kidney

Structure of Nephron

Nephron is the structural and functional unit of the kidney. There are over millions of nephrons in each kidney. There are two types of nephrons; they are cortical nephrons (reach short distance to the medulla) and juxta medullary nephrons (extend deep into the medulla).

Majority of the nephrons are cortical nephrons.

A nephron consists of a single long tubule and a ball of capillaries called the glomerulus. The tubule is closed at one end forming Bowman's capsule which surrounds the glomerulus. The other end of the tubule joins with the collecting duct.

Tubule consists of;

- Bowman's capsule
- Proximal convoluted tubule
- Loop of Henle
- Distal convoluted tubule

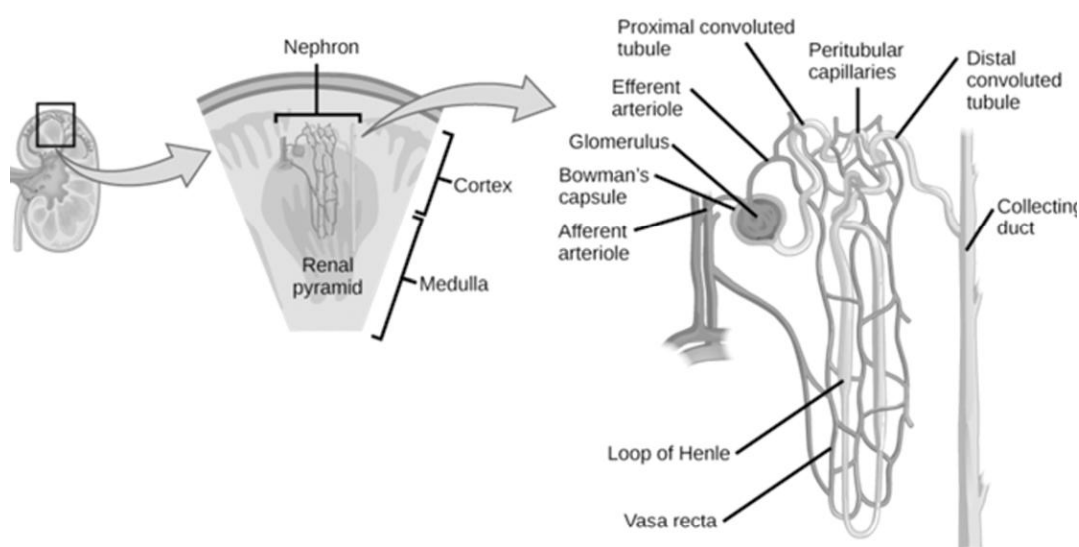


Fig 5.38: A nephron and associated blood vessels

Bowman's capsule (Glomerular capsule)

This is expanded and closed end of the tubular structure of the nephron. It is a cup shaped and double walled structure. Inner layer of Bowman's capsule consists of a single layer of flattened epithelial cells specialized for filtration. Outer layer of the Bowman's capsule composed of simple squamous epithelium. Space between the inner and outer layers is known as capsular space which is responsible for receiving of glomerular filtrate. The glomerular filtrate pass through three major tubular regions of the nephron: proximal convoluted tubule, loop of Henle and distal convoluted tubule.

Glomerulus

The glomerulus is a ball of capillaries which is surrounded by Bowman's capsule. The nephron is supplied with blood from the afferent arteriole. The blood vessel leaving away from the glomerulus is the efferent arteriole. The efferent arteriole has a smaller diameter than the afferent arteriole. This modification is important for increasing blood pressure in the glomerulus for ultrafiltration.

The efferent arteriole forms two capillary networks, one forms the peritubular capillaries which surrounds the proximal and distal convoluted tubules and the other network forms the vasa recta which extend towards the medulla surrounding the loop of Henle.

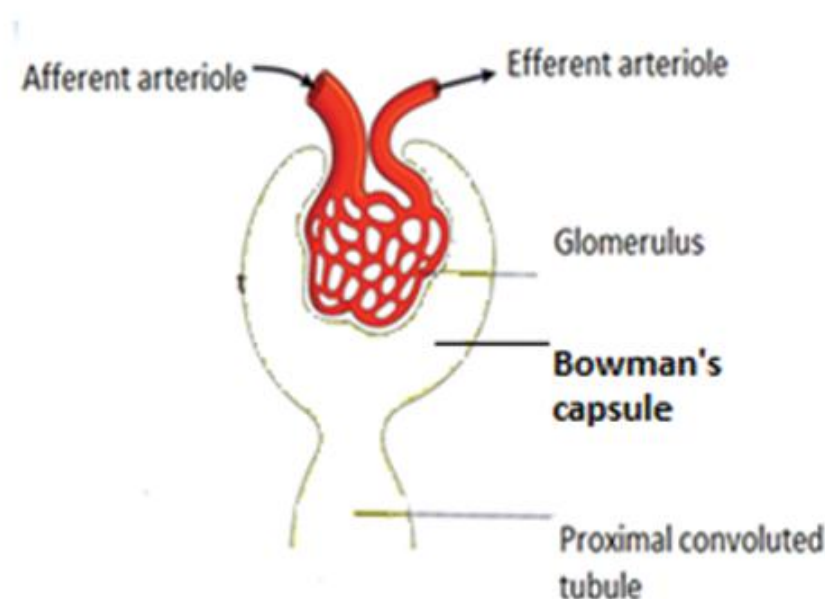


Fig 5.39 The Glomerulus and the Bowman's capsule

Proximal convoluted tubule

It is comparatively longer and wider than the distal convoluted tubule. This is lined by simple epithelium which has been specialized for selective reabsorption of substances (nutrients, ions and water) from the glomerular filtrate.

Loop of Henle

It is a 'U' shaped part of the nephron with descending limb and ascending limb which are lined by simple epithelium.

Lining of the descending limb of loop of Henle is specialized for water reabsorption as it allows free movement of water. But lining of the ascending limb of loop of Henle is impermeable to water.

Distal convoluted tubule

It is lined by simple epithelium which has been specialized for selective reabsorption of specific ions and water. It leads into collecting duct.

Main steps in urine formation

There are 3 processes involved in urine formation. They are;

- ultrafiltration
- selective reabsorption
- secretion

Ultra filtration

- Filtration of the blood under high pressure into the cavity of the Bowmans' capsule is called ultrafiltration.
- Filtration occurs through the capillary walls of glomerulus and inner wall of Bowman's capsule.
- Blood capillaries of glomerulus are porous and cells lining the Bowmans' capsule are specialized for filtration of small size molecules and ions. These specializations allow passage of water and small solutes through the blood capillary walls into the Bowman's capsule. But due to their large size, blood cells, platelets and large molecules such as plasma proteins do not pass into the Bowmans' capsule.
- The filtrate in the Bowmans' capsule contains salts, amino acids, glucose, vitamins, nitrogenous wastes and other small molecules. The composition of the glomerular filtrate is similar in composition to plasma with exceptions of blood cells, platelets and plasma proteins.

Selective reabsorption

- The process through which useful molecules, ions and water from the glomerular filtrate are recovered and returned to the interstitial fluid and then into capillary network of the tubules is called selective reabsorption.
- Most of the reabsorption from the glomerular filtrate back into the blood takes place in the convoluted tubule. Ions, water and valuable nutrients are reabsorbed either active or passive transport from initial filtrate.

Secretion

- The process by which foreign materials and substances not required to the body including waste are cleared from the peritubular capillaries and interstitial fluid into the filtrate is called secretion

- Secretion is required because such substances may not be entirely filtered due to the short time they remain in the glomerulus.
- Substances that are secreted in to the filtrate include H^+ , NH_3 , creatinine, drugs (e.g. penicillin, aspirin) and excess K^+ . Tubular secretion of H^+ and NH_3 are important to maintain the normal pH in the blood by formation of NH_4^+ in the urine. NH_3 can combine with H^+ to form NH_4^+
- Secretion occurs in the proximal and distal convoluted tubules. Secretion may be either active or passive depending on the location and / or the substance.

Process of urine formation

Glomerular filtrate in the Bowman's capsule which contains all the substances in the blood except blood cells, platelets and large molecules pass to the proximal convoluted tubule. At this region selective reabsorption of ions, water and valuable nutrients from the initial filtrate occurs. Nutrients especially glucose and amino acids are actively transported to the interstitial fluid. Cells lining the tubule actively transport Na^+ into the interstitial fluid and this transfer a positive charge out of the tubule drives the passive transport of Cl^- . Proximal tubule also reabsorbs K^+ and most of the HCO_3^- by passive transport. Reabsorption of HCO_3^- in the filtrate contributes to the pH balance in body fluids. As solutes move from the filtrate to interstitial fluid water is reabsorbed passively by osmosis. A major portion of water reabsorption from the filtrate occurs at this site. As the filtrate pass through the proximal convoluted tubule, secretion of specific substances into the filtrate takes place. Cells lining the tubule secrete H^+ (by active transport) and ammonia (by passive transport) into the lumen of the tubule. Secreted ammonia act as a buffer to trap H^+ forming NH_4^+ . In addition some materials such as drugs and toxins that have been metabolized in the liver are actively secreted into the lumen of the proximal convoluted tubule. As a result of water reabsorption and secretion of different substances, the filtrate becomes more concentrated as it passes through the proximal convoluted tubule.

As the filtrate moves into the descending limb of loop of Henle passive reabsorption of water through osmosis continues and filtrate becomes more concentrated. The filtrate reaches the ascending limb of the loop of Henle via the tip of the loop. Ascending limb is impermeable to water so that no water absorption takes place but a considerable amount of NaCl reabsorption occurs at this site. Most of the Na^+ is transported into the interstitial fluid by active transport. As a result of losing NaCl but not water the filtrate become more diluted as it moves towards the distal convoluted tubule.

The distal convoluted tubule plays an important role in regulating K^+ and NaCl concentration of body fluids. The amount of K^+ secreted (by active transport) into the filtrate and the amount of NaCl (by active transport) reabsorbed from the filtrate can be varied at this site according to the needs of the body. Distal tubule also contributes

to pH regulation by controlled secretion of H^+ and reabsorption of HCO_3^- . At the distal convoluted tubule passive water reabsorption can be increased under the influence of Antidiuretic hormone (ADH) to form concentrated urine. Aldosterone secreted by the adrenal gland stimulates increase reabsorption of Na^+ and water and excretion of K^+ . This filtrate of the distal convoluted tubule finally leads to the collecting duct.

As the filtrate pass along the collecting duct filtrate become concentrated and urine is formed. At this site water reabsorption can also be increased under the influence of ADH and urine become more concentrated. Aldosterone hormone stimulates active reabsorption of Na^+ and passive reabsorption of water at the collecting duct. Because of the high urea concentration in the filtrate at this region some urea diffuses into the interstitial fluid. Final processing of the filtrate at the collecting duct forms the urine.

When producing dilute urine the kidney actively reabsorb salts without allowing water to be reabsorbed by osmosis.

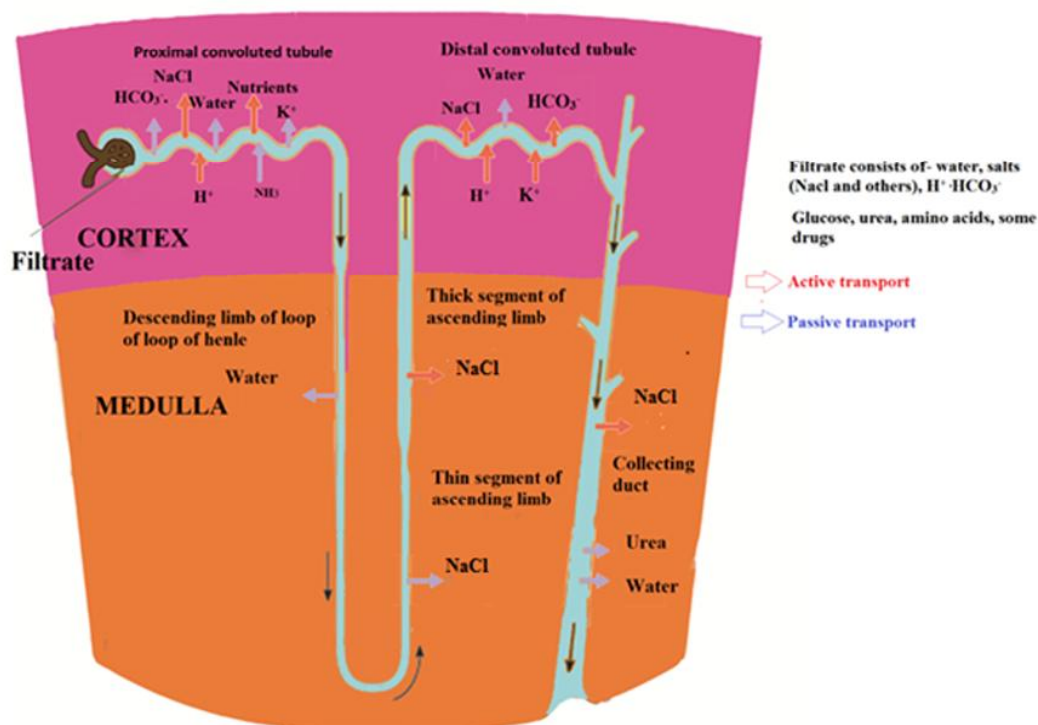


Fig 5.40 Reabsorption and secretion of different molecules and ions in the nephron and formation of urine in the collecting duct

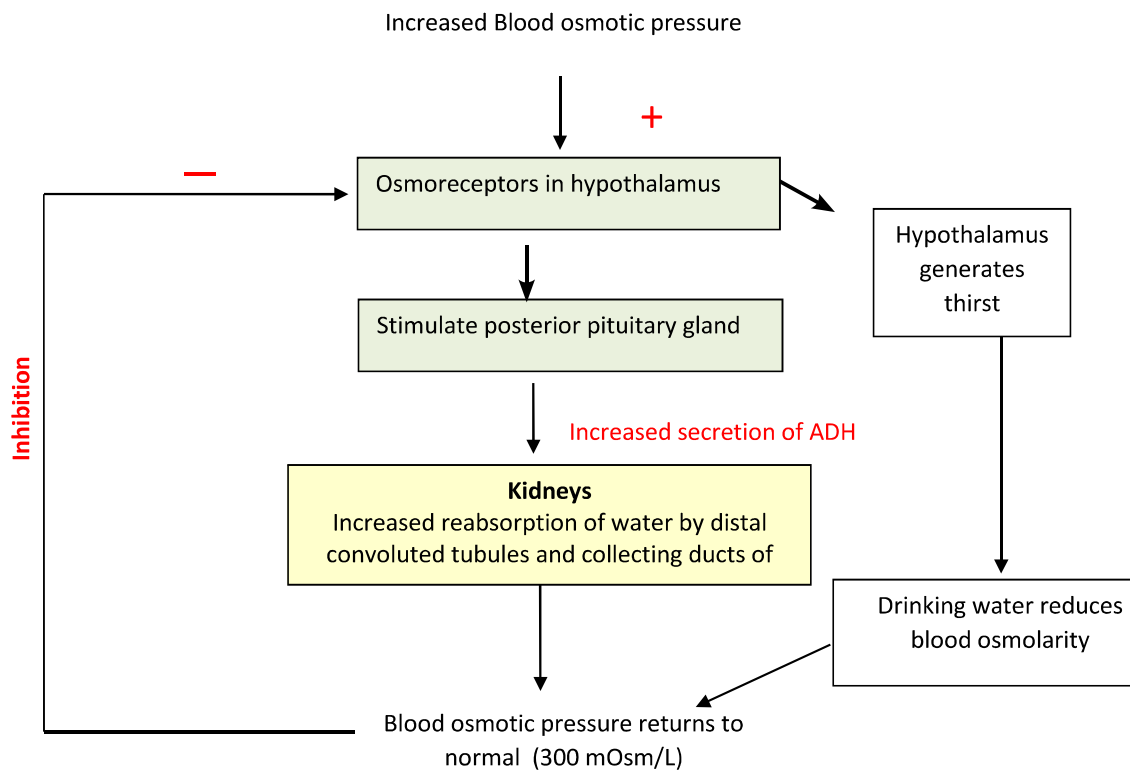
Role of hormones on the functions of the kidney

Fig5.41 Regulation of blood osmotic pressure and fluid retention in the kidney by ADH

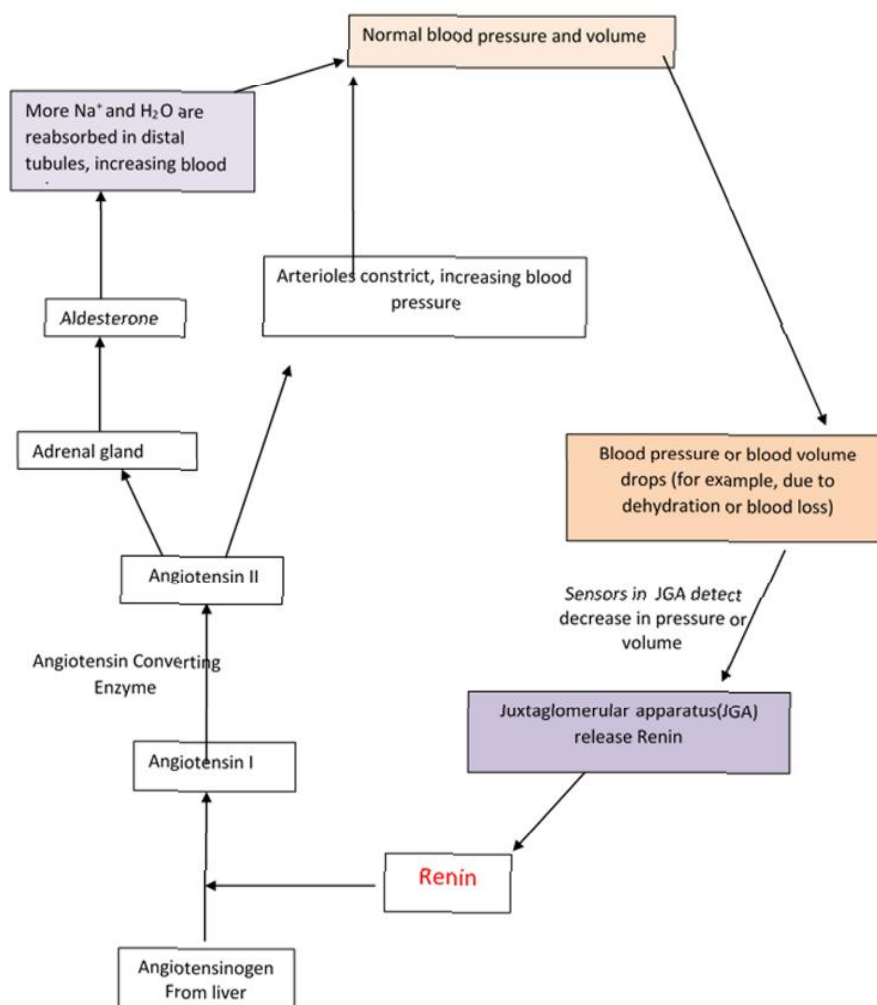


Fig5.42: Regulation of blood osmotic pressure and fluid retention in the kidney by ADH

Role of kidney in homeostasis

- Maintaining electrolyte and water balance in the body fluids (osmoregulation).
- Excretion of toxic waste products from the body.
- Regulating blood pH through acid base balance.
- Controlling blood volume and blood pressure.
- Secretion of erythropoietin hormone that stimulates red blood cell production.
- Production and Secretion of renin an enzyme important in control of blood pressure.

Disorders related to human urinary system**Bladder and kidney stones**

These are formed due to precipitation of urinary constituents (usually oxalates and phosphates) normally in urine. They are also called renal calculi.

Causes include

- Dehydration due to not drinking sufficient amount of fluids.
- Alkaline nature of urine.
- Infections that can alter pH of urine
- Metabolic condition.
- Family history.

Measures for prevention

- Drinking plenty of water

Kidney failure

This is due to the inability of the kidneys to function properly. Therefore waste products and excess fluid will be accumulated in the blood.

Reasons for kidney failure

- Diabetes
- High blood pressure
- Having family history
- Getting older

Chronic kidney disease (CKD)

It is a condition of gradual loss of kidney function over time. There are many reasons for Kidney failure:

- Diabetes
- High blood pressure

- Having family history
- Getting older

Prevent CKD

Follow a low salt, low fat diet

Doing proper exercise

Having regular check ups

Prevent smoking

Dialysis

Dialysis is done for the patients with kidney failure. It is a process of removing excretory products, excess solutes and toxins from the blood by an artificial method.

Chronic kidney disease of unknown/uncertain etiology (CKDu) in Sri Lanka

- It is a condition of gradual loss of kidney function over time. The root cause of CKDu has not been definitively established yet – hence it is referred to as ‘Chronic kidney disease of unknown/uncertain etiology’. However it is a different form of chronic kidney disease (CKD), which is associated with conventional risk factors such as diabetes and high blood pressure, genetic disorders and urinary tract problems
-
- The onset of the disease appears to be asymptomatic, and by the time the patient seeks treatment the kidneys have reached a stage of irreversible damage -end stage renal disease (ESRD).
- In Sri Lanka, initially CKDu was prevalent among rural communities in North Central province (Medawachchiya, Kabithigollawa, Padaviya, Medirigiriya, regions), Uva (Girandurukotte), Eastern Provinces (Dehiattakandiya). CKDu is also reported in North Western, Southern and Central provinces, and parts of the Northern Province of the island.

Hypothesized reasons for CKDu

Cause of CKDu seems to be multifactorial.

- Exposure to Heavy metal/ metalloid such as Arsenic (As) and Cadmium (Cd) through food and water.
- Usage of low quality utensils for preparation of foods.
- High Flouride (F) levels in water.
- Exposure to pesticides.
- Genetic factors.
- Malnutrition and dehydration.

Notes:

This is to acknowledge that some of the diagrams used in this book have been taken from various electronic sources using internet . This book is not published to make profit and sold only to cover cost.

The resource book is prepared according to the subject content and learning outcomes of the G.C.E. (A.L) Biology new syllabus which is implemented from 2017.

The content of this Resource book declares the limitation of the G.C.E. (A.L) Biology new syllabus which is implemented from 2017.

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05

Animal Form and Function**Processes and systems involved in coordination**

Coordination between stimuli and responses is needed to maintain constant internal environment inside the body of an organism for existence.

Systems contributing to coordination

Animals unlike plants have two different but related systems for coordination of body function. They are the nervous system and the endocrine system.

Table 5.1: Similarities and differences (in relation to coordination) of the nervous system and endocrine system

Feature	Nervous coordination	Hormonal coordination
Transmission	through neurons	through blood
Nature of transmitter	chemical and electrical	chemical
Response	localized	diffused
Time taken to start the response	fast acting	slower action
Duration of response	short	long

Organization of nervous systems in different animal Phyla

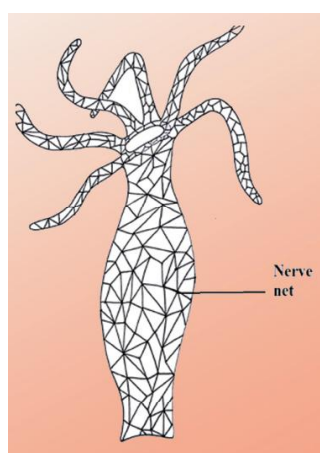
Animals have specialized systems of neurons to sense their surroundings and respond rapidly. In the animal kingdom, cnidarians are the simplest animals having a nervous system. They have a diffuse nerve net which is composed of interconnected individual neurons.

In more complex animals, the nervous systems contain groups of neurons organized into nerves, and often ganglia and a brain. In some platyhelminthes such as Planaria, the nervous system contains a pair of ganglia in the anterior region (brain) and a pair of ventral nerve cords that runs longitudinally. In planarians, the eye spots which are located near the ganglia act as photoreceptors. Annelids and arthropods have a somewhat complicated brains and ventral nerve cords. The ventral nerve cord contains ganglia. They are segmentally arranged. Nervous system

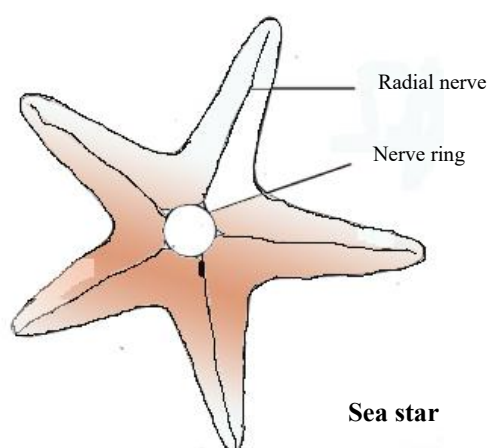
of echinodermites is composed of radial nerves and a nerve ring. Nervous system of the chordates consists of a central nervous system (CNS) and a peripheral nervous system (PNS). The CNS is composed of the brain and the spinal cord. The PNS is composed of nerves and ganglia.

Table 5.2: Different organism phyla and their nervous organization

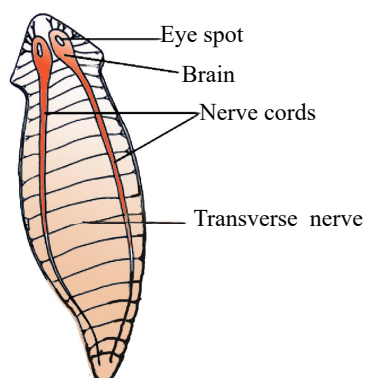
Phylum	Organization	Example
Cnidaria	Nerve net	Hydra
Platyhelminthes	Brain, longitudinal nerve cords	Planaria
Annelida	Brain, ventral nerve cord, segmental ganglia	Leech
Arthropoda	Brain, Ventral nerve cord, Segmental ganglia	Cockroach
Echinodermata	Nerve ring and radial nerves	Sea star
Chordata	Brain, spinal cord (dorsal nerve cord), nerves and ganglia	Gecko



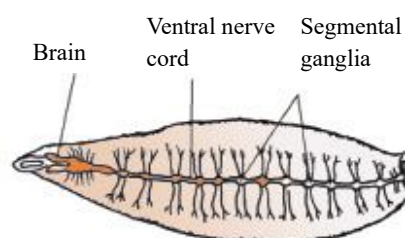
Hydra



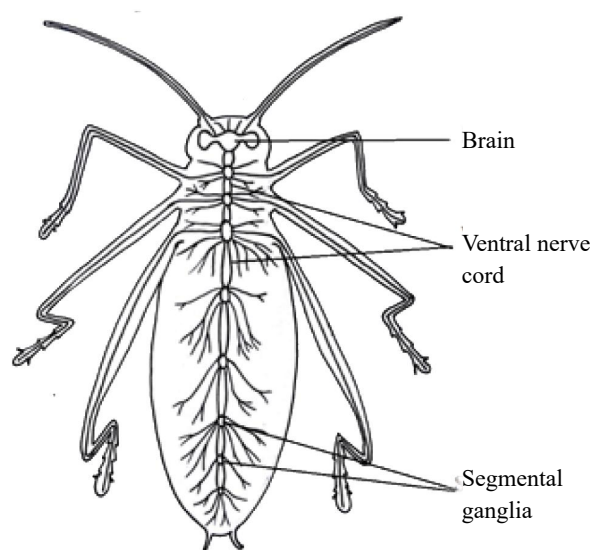
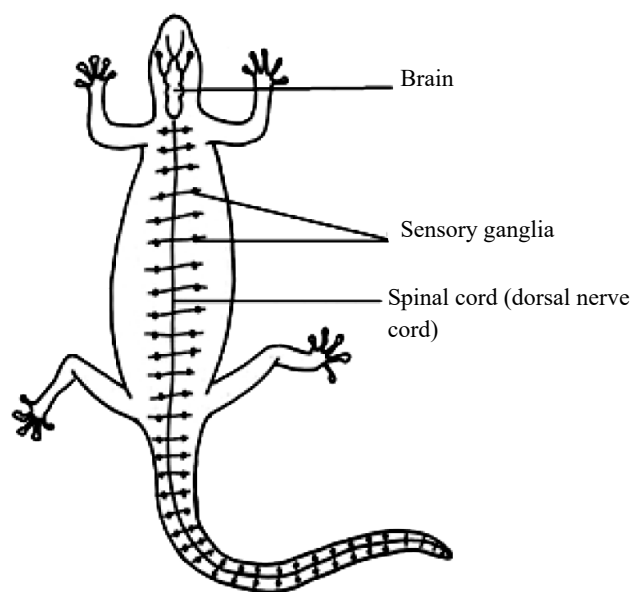
Sea star



Planaria



Leech

**Insect****Gecko****Figure 5.1: Organization of nervous systems in different animal phyla**

The gross structure and the functions of the human nervous system

Organization and main parts of the human nervous system

Human nervous system consists of central and peripheral nervous systems. In vertebrates, the brain and the spinal cord form the central nervous system. Nerves and ganglia forms the main components of the peripheral nervous system.

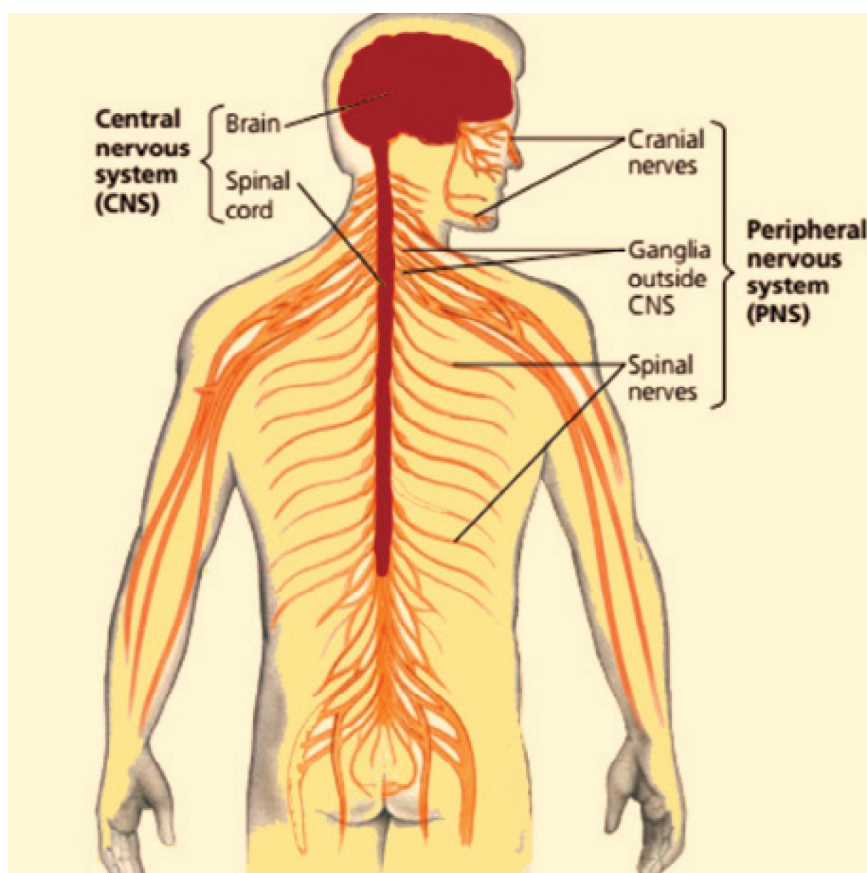
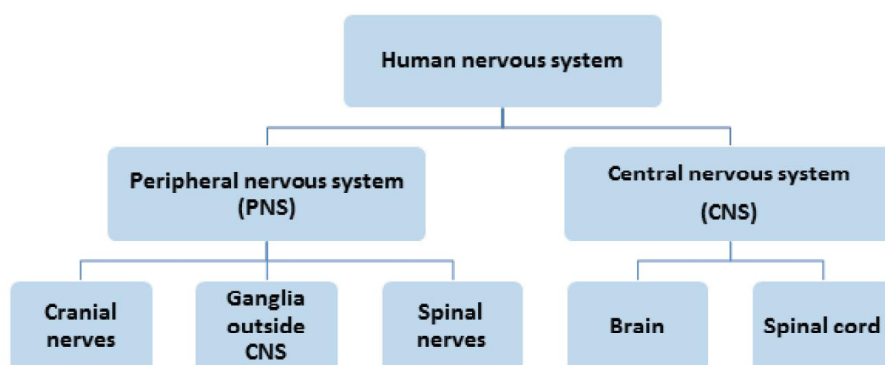


Figure 5.2: The organization of human nervous system

Central nervous system (CNS)

Central nervous system consists of the brain and the spinal cord. In vertebrates, the CNS develops from the hollow dorsal nerve cord during embryonic development. Anterior part of the central nervous system enlarges and forms the brain which has three major regions: forebrain, midbrain and hindbrain. The central canal in the brain forms four irregular shaped cavities called ventricles. The brain contains four ventricles: three ventricles are present in the fore brain and one ventricle is in the hind brain. This central canal continues in the spinal cord. The ventricles and central canal contains cerebrospinal fluid. This fluid helps to maintain uniform pressure within the CNS and act as a shock absorber between the brain and skull. It also helps to circulates nutrients and hormones as well as to remove waste products.

The brain and the spinal cord have several adaptations to be protected from physical injuries. The brain is enclosed by a skull. The spinal cord is surrounded by vertebrae which forms the vertebral column. Further protection to the CNS is given by three layers of tissues called the meninges. The outermost layer is called the dura mater, the innermost layer as pia mater and in between these two layers is the arachnoid mater.

Main parts of the human Brain

The forebrain, midbrain and hindbrain of the human embryo develops into the adult brain. The forebrain gives rise to the cerebrum, thalamus, hypothalamus and pineal body. The mid brain gives rise to part of the brain stem. The hind brain gives rise to cerebellum, pons varoli and medulla oblongata. The brain stem consists of the midbrain, pons Varolii and medulla oblongata.

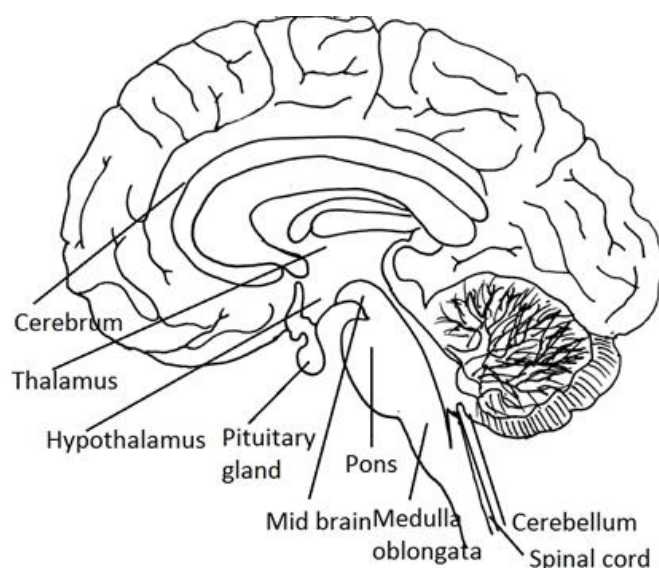


Figure 5.3 : The longitudinal view of the human brain

Cerebrum

Cerebrum is the largest part of the human brain. It is divided by a deep cleft into right and left cerebral hemispheres. The superficial part of the cerebrum is composed of nerve cell bodies (or grey matter) forming the cerebral cortex and deeper layers consist of nerve fibers (or white matter). The two cerebral hemispheres are connected by corpus callosum which is a mass of white matter. The cerebral cortex shows many infoldings to increase the surface area of the cerebrum. The cortex of each cerebral hemisphere is divided into four lobes: frontal lobe, temporal lobe, parietal lobe and occipital lobe.

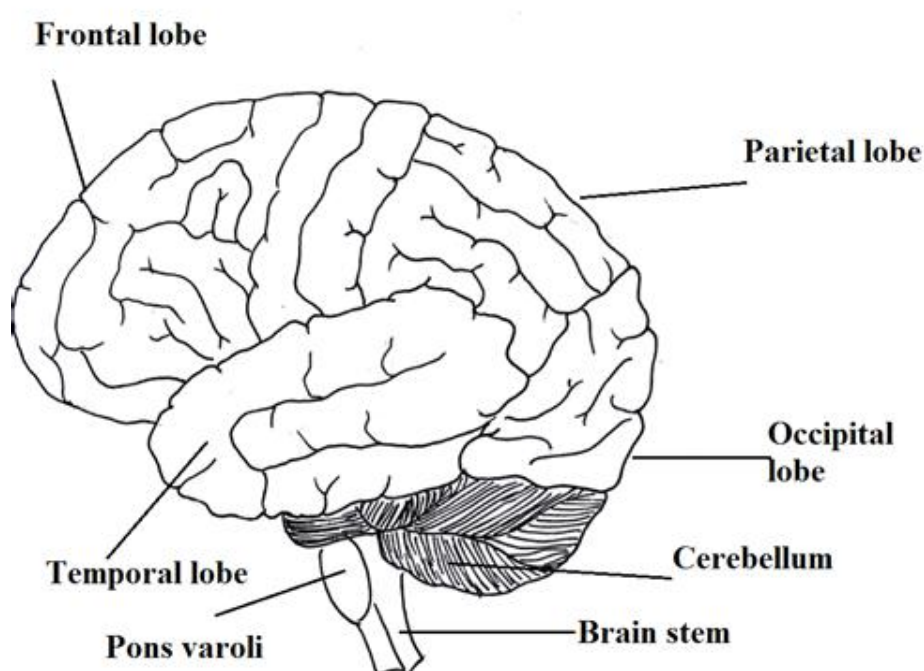


Figure 5.4 : The human cerebral cortex

Three main functional areas of the cerebral cortex have been identified. They are;

- Sensory areas which receive and process sensory information including the perception of pain, temperature, touch, sight, hearing, taste and smell
- Association areas which are responsible for recognition and interpretation of sensory information and integration and processing of complex mental functions such as memory, intelligence, reasoning, judgment and emotions
- Motor areas which are responsible for directing skeletal (voluntary) muscle movement through the initiation and control of voluntary muscle contraction

Thalamus

Thalamus is situated within the cerebral hemispheres just below the corpus callosum. It consists of two masses comprising grey and white matter.

Functions:

Thalamus acts as the main input centre of sensory information from special sense organs and sensory receptors in the skin and integral organ. This sensory information is sorted and directed to specific location of the cerebral cortex for further processing and perception. The thalamus relays and redistributes nerve impulses from most parts of the brain to cerebral cortex.

Hypothalamus

Hypothalamus is situated below and in front of the thalamus, immediately above the pituitary gland. It is linked to the posterior lobe of the pituitary gland by nerve fibers and to the anterior lobe by a complex system of blood vessels.

Functions:

- Regulates body temperature
- Regulates thirst and water balance
- Regulates appetite
- Regulate sleep and wake cycles
- Control of autonomic nervous system
- Initiates fight-or-flight response
- Source for posterior pituitary hormones and releasing hormones that act on anterior pituitary.
- Plays a role in sexual behaviours

Mid brain

Mid brain is the upper part of the brain stem. It is situated between the cerebrum above and the pons below surrounding the cerebrospinal fluid filled connection of the third and fourth ventricles. Mid brain contains aggregates of nerve cell bodies and nerve tracts which connect the cerebrum with lower brain and spinal cord.

Functions:

- Acts as relay stations for ascending and descending nerve fibers
- Receives and integrates sensory information (auditory and visual) and sends it to particular regions of the forebrain, coordinates auditory and visual reflexes

Pons Varolii

Pons Varolii, (a part of the brain stem) is located in front of the cerebellum, below the mid brain and above the medulla oblongata. It contains nerve fibers that form a bridge between the two hemispheres of the cerebellum. It also contain nerve fibers passing between higher levels of brain

and spinal cord. Groups of nerve cell bodies in the pons form centers that regulate respiration. Some nerve cell bodies in the pons act as relay stations.

Functions:

- Transfers information between PNS and the midbrain and forebrain
- Coordinates large scale body movements such as climbing and running
- Together with the medulla oblongata helps regulate respiration.

Medulla oblongata

Medulla oblongata is the lowest part of the brain stem which extends from the pons above and is continuous with the spinal cord below. It consists of cardiovascular centre, respiratory center and reflex centers.

Functions:

- Transfers information between PNS and the mid brain and the fore brain
- Coordinates various body movements such as running, climbing
- Controls several autonomic, homeostatic functions including breathing, heart and blood vessel activities (contains respiratory centre, cardiovascular centre)
- Controls involuntary reflexes such as vomiting, swallowing, coughing, sneezing through reflex centres

Cerebellum

Cerebellum is located behind the pons Varolii and below the posterior portion of the cerebrum. It is also made up of two hemispheres.

Functions

- Coordinates voluntary muscular movements
- Maintains posture and balance
- Helps in learning and remembering motor skills

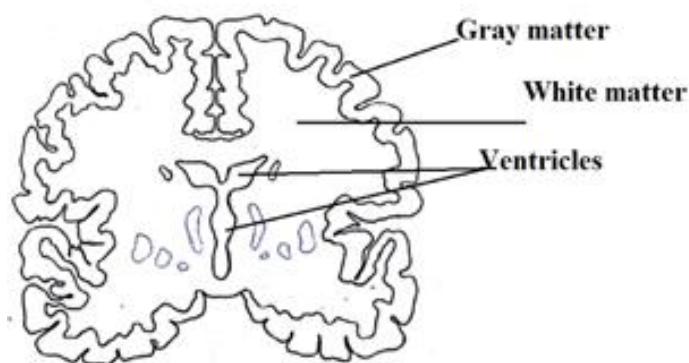


Figure 5.5 : T.S of the human brain

Spinal cord

The spinal cord is an elongated cylindrical structure suspended in the vertebral canal. It is continuous with the medulla oblongata. Centre of the spinal cord contains the central canal which is surrounded by grey matter. Outer region of the spinal cord is made up of white matter.

Functions:

- Links the central nervous systems to sensory and motor neurons and facilitates nerve impulse propagation towards the brain and away the brain
- Coordinates and produces reflexes

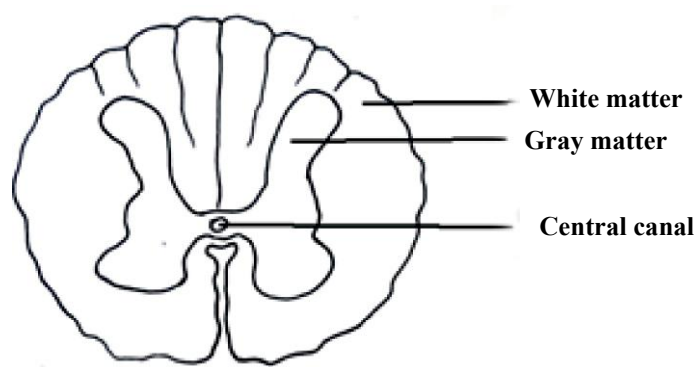


Figure 5.6: T.S of the spinal cord

Peripheral nervous system (PNS)

Peripheral nervous system is made up of cranial nerves, spinal nerves and autonomic nervous system (with ganglia). It transmits impulses to and from CNS regulating both an animal's movement and its internal environment.

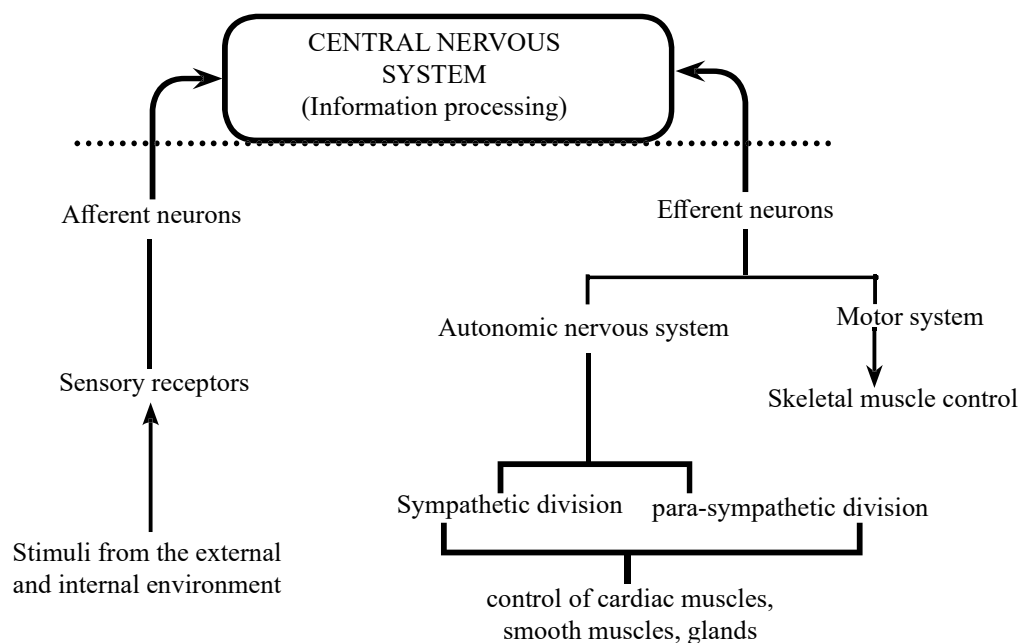


Figure 5.7 : Peripheral nervous system of a vertebrate (Functional hierarchy)

Sensory information from sensory receptors reaches the CNS along PNS neurons referred to as afferent neurons (sensory neurons). Within the CNS this information is processed and instructions are transmitted to effector tissues/organs (muscles, glands and endocrine cells) along PNS neurons referred to as efferent neurons (motor neurons).

PNS consists of two efferent components;

- The motor system- It consists of neurons that carry nerve impulses to skeletal muscles. So it controls voluntary activities.
- Autonomic nervous system- It generally controls the involuntary activities of the body. Autonomic nervous system consists of neurons which carry impulses to control activities of smooth muscles, cardiac muscles and glands.

Autonomic nervous system consists mainly of two divisions:

- Sympathetic division
- Parasympathetic division

Sympathetic and parasympathetic nervous system

The majority of the body organs are supplied by both sympathetic and parasympathetic nerves which have antagonistic (opposite) functions. Sympathetic stimulations prepare the body to deal with exciting/ stressful situations and energy generating situations (fight- or -flight). Parasympathetic division causes opposite responses that promote calming or a return to self-maintenance functions (rest and digest).

The two divisions differ in overall functions, organization and the signal released. Parasympathetic nerves exit the CNS at the base of the brain or the spinal cord as cranial nerves or spinal nerves respectively. On the other hand sympathetic nerves exit only from the spinal cord. Different neurotransmitters enable the two systems to bring about two opposite effects in different organs such as lung, heart, intestine and bladder. For example, when the neurotransmitter secreted by the parasympathetic division is acetylcholine, the sympathetic division secretes norepinephrine.

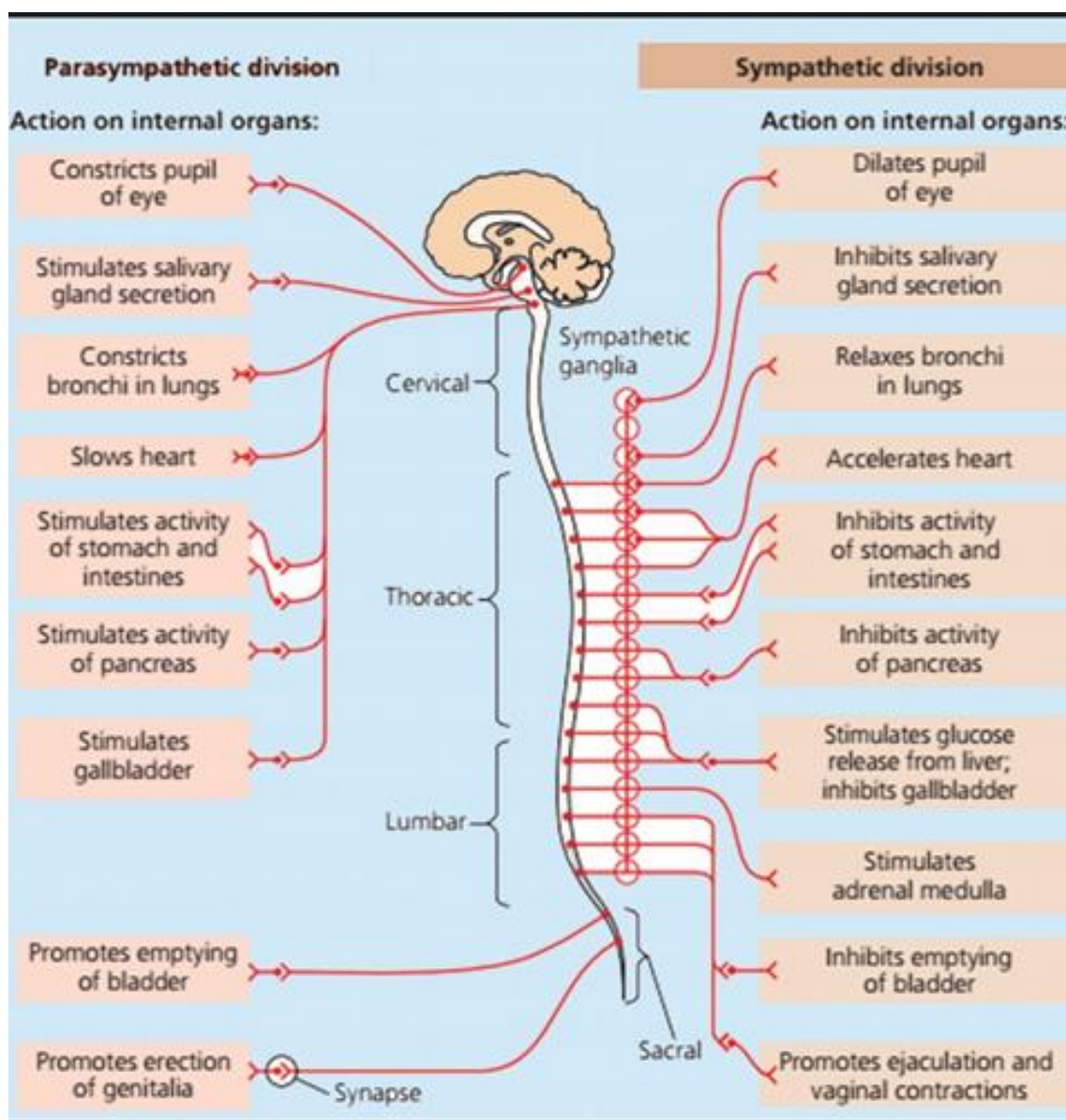


Figure 5.8: The autonomic nervous system (Parasympathetic and sympathetic division)

How nerve impulses are generated and transmitted

In all cells including neurons, ions are distributed unequally between the cell interior and exterior (extra cellular fluid). Generally the inside of the cell is negatively charged whereas the exterior is positively charged. These opposite charges are attracted across the plasma membrane and as a result it creates a voltage difference across the membrane that is referred to as membrane potential.

Resting potential

When a neuron is at rest (when not sending a signal/non conducting), the membrane potential is called the resting potential. In a non-conducting neuron the resting potential is typically between -60 mV and -80 mV.

The resting membrane potential is maintained by;

- Distribution of ion concentrations inside and outside of the neuron- In a non conducting neuron the concentration of K^+ is higher inside the cell while concentration of Na^+ is higher outside. In addition there are Cl^- and other large anions (proteins) inside the cell. As a result a neuron has a negative charge inside the cell and positive charge outside the cell.
- Selective permeability of the plasma membrane to K^+ and Na^+ ions- There are potassium channels and sodium channels, which are membrane bound proteins that are able to open or close in response to stimuli. Potassium channels allows only K^+ ions to pass whereas sodium channels allow only Na^+ to pass. These channels allow K^+ and Na^+ to diffuse according to their concentration gradient. However there are more potassium channels open than sodium channels. As a result there is a net negative charge inside the cell.
- Sodium-potassium pump- This pump helps to maintain Na^+ and K^+ gradient across the membrane by transporting three Na^+ out of the cell for every two K^+ that it transports in. This pump uses ATP to actively transport these ions.

Action potential

An action potential occurs due to a change in membrane potential above a threshold value due to a stimulus. The action potential has the following phases: depolarization, repolarization and hyperpolarization.

Depolarization: A change in the cell's membrane potential such that the inside of the membrane is made less negative relative to the outside. Depolarization results due to Na^+ inflow in response to a stimulus.

Repolarization: Sodium channels close blocking Na^+ inflow. However most potassium channels open permitting K^+ outflow. This makes the inside of the cell negative.

Hyperpolarization: Sodium channels are closed but potassium channels are opened. As a result the inside of the membrane is more negative.

Refractory period

Refractory period is the short time immediately after an action potential in which the neuron cannot respond to another stimulus, owing to the inactivation of sodium channels. This prevents the reverse conduction of an impulse in an axon.

Generation of action potential

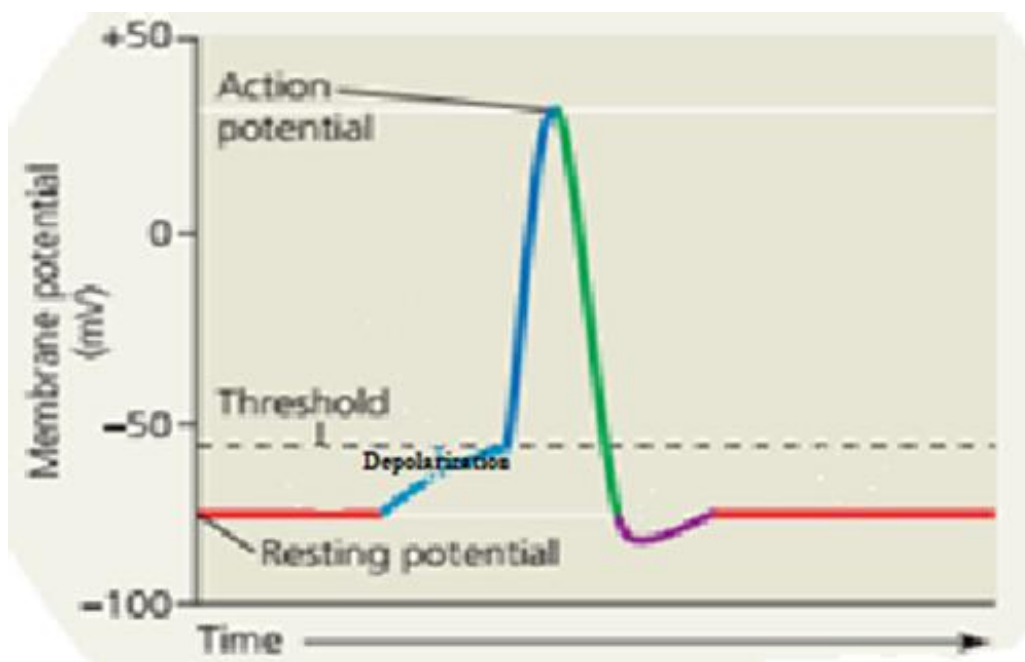


Figure 5.9: Graph of generating action potential

Conduction of action potential (nerve impulse)

- A series of action potentials that move along an axon is defined as a nerve impulse.
- An action potential is generated due to Na^+ inflow (depolarization) at one location in the axon.
- This axon potential spreads to the neighboring location while the initial location repolarizes.
- This depolarization-repolarization process is repeated through the axon.

The speed of conduction depends on:

Diameter of the axon- The conduction speed increases with the increase in axon diameter.

Presence of myelinated axon (in myelinated neuron, axon potential jumps from one node of Ranvier to the next)

Synapses

A synapse is the junction where a neuron (presynaptic cell) communicates with another cell (postsynaptic cell) across a narrow gap (synaptic cleft). Postsynaptic cell may be another neuron, muscle cell or secretory cell. This junction where one neuron communicates with the next cell using a chemical (neurotransmitter) is called a chemical synapse. Some neurons can also communicate through direct electrical connections (electrical synapse).

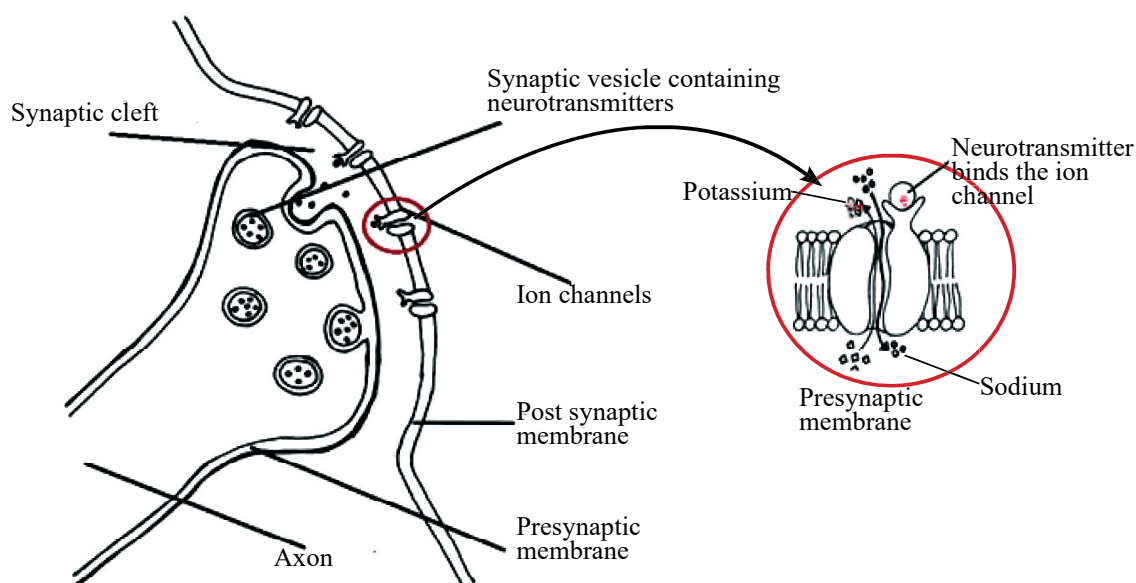


Figure 5.10: A synapse which communicates through a neurotransmitter

Mechanism of transmission of nerve impulses through chemical synapses

- An action potential at an axon terminal depolarizes the plasma membrane of presynaptic cell.
- Depolarization at the presynaptic terminal causes Ca^{2+} to diffuse into the terminals.
- The rise in Ca^{2+} causes binding of synaptic vesicles containing neurotransmitters to the presynaptic membrane.
- This results in the release of the neurotransmitters into the synaptic cleft.
- Neurotransmitters diffuse across the synaptic cleft.
- Neurotransmitters bind and activates specific receptors in the postsynaptic cell membrane.
- If acetylcholine is taken for example, the binding of neurotransmitters to the post synaptic membrane allows Na^+ and K^+ to diffuse across the post synaptic membrane.
- Depolarization takes place in the post synaptic membrane and it reaches the action potential
- After passing the nerve impulse to the postsynaptic cell, the signal is terminated either by:
 - Enzymatic hydrolysis of neurotransmitters
 - Recapture of neurotransmitter into the presynaptic terminals.

Neurotransmitters

Neurotransmitters are the molecules that are released from the synaptic terminals of presynaptic neuron and diffuse across the synaptic cleft, bind to the receptors at the postsynaptic membrane, triggering a response.

Common neurotransmitters are;